How the RREA Model Is Built on WarpX/AMReX: A Beginner-Friendly Architecture Guide

RREA WarpX/AMReX project

Abstract

This guide explains, from first principles, how the RREA (Relativistic Runaway Electron Avalanche) and **relativistic feedback** simulation model is constructed on top of two open-source frameworks: **AMReX** (a block-structured mesh and particle library) and **WarpX** (a particle-in-cell plasma code built on AMReX). The model tracks not only the forward electron avalanches but also the backward-propagating bremsstrahlung photons and pair-produced positrons that can re-seed new avalanches at the source region: Dwyer's relativistic feedback mechanism, which turns isolated avalanches into a self-sustaining discharge and is essential for Terrestrial Gamma-ray Flash physics. The guide is written for a reader who knows basic plasma/atmospheric physics and some C++, but who has never used WarpX or AMReX. It describes what the frameworks provide out of the box, exactly where and how the RREA code attaches to them, what happens during one simulation time step, and why the project is organized the way it is. It documents the schema-6 transport and Maxwell TM engine.

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How to read this guide. If you are new to the project, read Sections 1–2 for the physics goal and a five-minute tour of AMReX and WarpX, then Sections 3–5 for how the code attaches to WarpX and what the input deck controls. The heart of the guide is the walkthrough of one simulation time step (Section 6); nearly every later section expands one box of its data-flow figure. The deeper dives (the interaction tables (Section 7), the avalanche-length benchmark (Section 12), and the fluid and field solve (Section 8)) can be read independently, as can the diagnostics and observable-signature sections. Recurring terms are collected in the Glossary at the end.

1 What the Model Simulates

A thundercloud can sustain quasi-static electric fields above the *runaway breakeven* threshold. In such fields, an energetic electron (MeV scale) gains more energy from the field than it loses to air, so it "runs away," and its hard collisions knock out further energetic electrons: a Relativistic Runaway Electron Avalanche. The avalanche emits bremsstrahlung gamma rays (the mechanism behind Terrestrial Gamma-ray Flashes), creates positrons, and, crucially, also deposits an enormous number of *low-energy* electrons and ions that make the air conductive. That conductivity discharges the very field driving the avalanche, so the system is self-limiting.

On top of the single-avalanche picture sits **relativistic feedback** (Dwyer): some of the avalanche's bremsstrahlung photons Compton-backscatter or propagate upstream, and some of its pair-produced *positrons* (whose charge is positive) are accelerated *backward* along the field. When such a backward-moving photon or positron reaches the region where the avalanches start, it can create a fresh energetic electron there (by Compton scattering / photoelectric absorption, or by the positron's hard Bhabha collisions), seeding a brand-new avalanche. If, on average, each avalanche seeds more than one successor, the discharge becomes self-sustaining and grows explosively until the field collapses. Whether a thundercloud crosses that feedback threshold is one of the central physics questions this model exists to answer, which is why photons and positrons are tracked as full kinetic species rather than treated as passive losses (Figure 1).

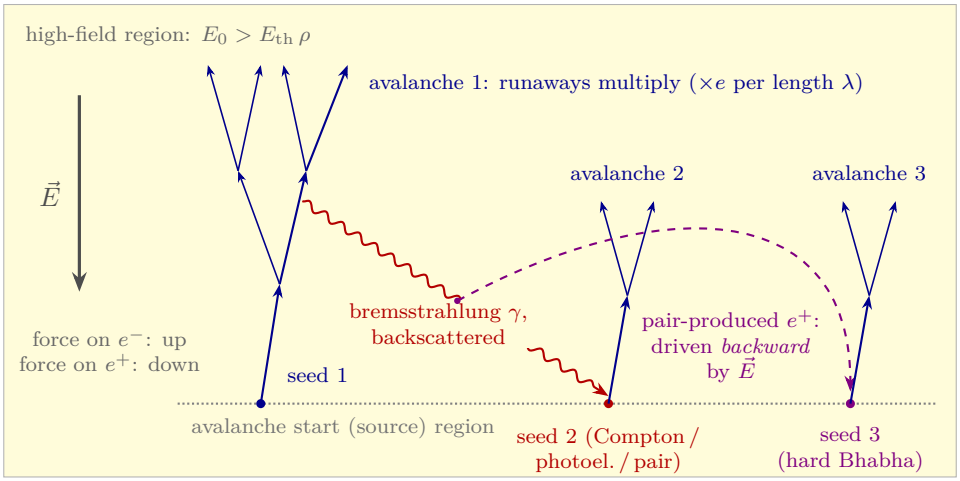


Figure 1: The relativistic feedback loop (Dwyer) the model is built to capture. A seed electron runs away and multiplies into avalanche 1. Some of its bremsstrahlung photons propagate or Compton-backscatter *against* the avalanche direction and liberate fresh energetic electrons near the start region; positrons pair-produced by those photons carry positive charge and are therefore accelerated *backward* by the same field, seeding further electrons through hard Bhabha collisions. If each avalanche seeds on average at least one successor, the discharge is self-sustaining.

Modeling this therefore needs three coupled ingredients:

1. **High-energy particle transport** (electrons, photons, positrons, $\gtrsim$ keV–MeV–GeV): individual particles pushed through air with Monte Carlo collisions. Photons and positrons get the same first-class treatment as electrons precisely because they carry the relativistic feedback.
2. **A low-energy plasma fluid** (electrons below a cutoff, positive and negative ions): far too numerous to track as particles; represented as densities on the mesh, evolving by attachment, recombination, and drift, and producing a conductivity σ.
3. **Self-consistent field feedback**: the electric field is the imposed thundercloud profile *relaxed* by the conductivity and by the space charge of both populations.

The RREA model implements (1) as a WarpX particle extension, (2) and (3) as a native AMReX library, and couples them inside the WarpX time step. The simulation geometry is cylindrical RZ (radius r, altitude z), which matches the natural symmetry of a vertical avalanche column.

2 The Building Blocks: AMReX and WarpX in Five Minutes

2.1 AMReX

AMReX is a C++ framework for block-structured mesh codes. The concepts used constantly in this project:

- **Box / BoxArray**: the rectangular index space of the mesh is chopped into boxes (here up to 64×64 cells, set by `amr.max_grid_size`). Boxes are the unit of parallel distribution.
- **DistributionMapping**: assigns each box to one MPI rank.
- **MultiFab**: "multiple FArrayBoxes" (Fortran array boxes), *the* AMReX data structure: one floating-point field (e.g. electron density) stored box-by-box across all ranks, with ghost cells for stencils. The fluid, charge, and WarpX-facing field arrays are MultiFabs; the Maxwell update uses a replicated staggered state assembled from them.
- **ParticleContainer**: distributed particle storage; each particle carries position, momentum, a statistical *weight* (how many real particles one computational "macroparticle" represents), and optional extra components.

2.2 WarpX

WarpX is an electromagnetic/electrostatic particle-in-cell (PIC) code built on AMReX. A PIC code advances plasma in a loop: deposit particle charge onto the mesh → solve for fields → interpolate ("gather") fields to particles → push particles. WarpX provides, ready-made:

- the RZ cylindrical geometry and distributed particle storage;
- named particle species defined in a text input deck (`particles.species_names = ...`), each backed by a ParticleContainer;
- the main time loop (`WarpXEvolve.cpp`), field containers, I/O, checkpointing scaffolding, and MPI orchestration;
- an electrostatic solver, `ComputeSpaceChargeField`, which the RREA model *replaces* (Section 4).

The project gitlinks pin **WarpX 26.06** and **AMReX 26.06** in protected archival forks. AMReX is built from source inside the WarpX build; the executable is RZ and double precision, with MPI and CPU/OpenMP or CUDA execution (Section 14).

The key idea to keep in mind: *WarpX supplies particles, mesh, and the time loop; the RREA code supplies the physics of air and the field solver, and inserts itself at a few precisely chosen points of the loop.*

3 The Big Picture: Overlay + Native Library

The protected forks archive pristine release commits; this repository carries no modified copy of either framework. The RREA implementation consists of two parts:

Native library A standalone static library (`rrea_amrex_field_solver`; sources in `native_amrex_rrea/src` and `native_amrex_rrea/include/rrea/`) that depends only on AMReX, not on WarpX. It contains the fluid state, the Maxwell advance, the field initializer and atmosphere profile, the interaction-table loader, the seed schedule, and the event accumulator. Because it has no WarpX dependency, this physics can be built and tested without constructing WarpX. Focused MPI variants additionally exercise distributed invariants.

WarpX overlay (`native_amrex_rrea/warpx_overlay/Source/Rrea`) The WarpX-aware kinetic-transport and coupling layer. It owns the particle loops and their shared final-state primitives, links the native library, and is copied into a WarpX source tree at build time.

Native library (AMReX only)	**Overlay (WarpX-aware transport/coupling)**
`LowEnergyFluidState`	`RreaParticleInteraction` (MC engine)
`RreaAmrexAdvance`	`TransportStep` (three species phases + commit)
`RreaFieldInitializer`, `RreaAltitudeProfile`	`RreaRng` (counter-based RNG)
`RreaInteractionTables` (schema-6 loader)	checkpoint rank-state helpers
`RreaEventAccumulator`, `RreaSeedSchedule`	

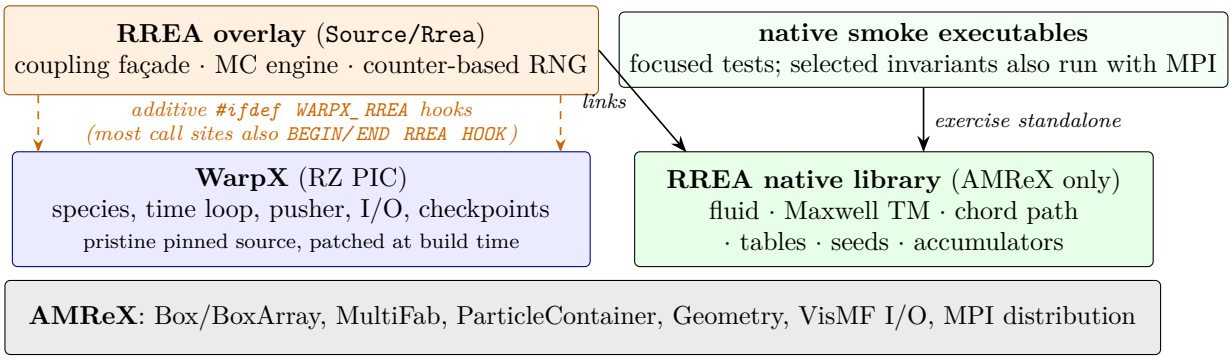


Figure 2: Where each piece of code sits. WarpX and the RREA native library both stand directly on AMReX; the overlay is the only WarpX-aware code and is grafted onto a pristine WarpX tree at build time through guarded, exact-match, idempotent patches (Section 4). The native library is exercised standalone by the native smoke executables before it ever runs inside WarpX; the production executable `warpx.rz` links all of it into one binary.

This split follows dependency ownership (Figure 2): AMReX-only mesh, field and table code lives in the native library; WarpX particle transport and coupling live in the overlay. Header primitives keep much of the transport physics independently testable without running WarpX.

4 How the Code Attaches to WarpX

4.1 The source-preparation step

A pristine WarpX checkout is never edited by hand. The script `scripts/prepare_rrea_warpx_source.py` builds the combined source tree:

1. copies the pristine WarpX tree to a build location (it refuses to run in-place);
2. copies the overlay directory in as `Source/Rrea`;
3. applies a small set of guarded patches to stock WarpX files. Every insertion is `#ifdef WARPX_RREA`-guarded; the seed-injection, field-solve and checkpoint call sites are additionally wrapped in `// BEGIN RREA HOOK ... // END RREA HOOK` comments. Each patch is applied by an exact-match text search ("needle"). The patch operations are idempotent, so `--refresh` can update an existing prepared tree; if a WarpX update moves a needle, the script *aborts* instead of guessing.

The patches add guarded declarations and calls without deleting stock behavior. The full list of patched files:

WarpX file	What the patch adds
`CMakeLists.txt`	`add_subdirectory(Source/Rrea)`; overlay sources join the WarpX library target with `WARPX_RREA=1`.
`Source/WarpX.H`	a `SpaceChargeSolvePurpose` enum and a `purpose` parameter on `ComputeSpaceChargeField`, so the hook can distinguish initialization solves from evolution solves.
`Source/FieldSolver/ WarpXSolveFieldsES.cpp`	**the field-solver replacement**: just before the electrostatic solver call in `ComputeSpaceChargeField` (after its optional field reset), call `RreaWarpXCoupling::BeforeFieldSolve(...)`; if it returns true, WarpX's own electrostatic solve is skipped entirely.
`Source/Evolve/WarpXEvolve.cpp`	three calls in the step loop: `InjectScheduledSeeds` (before the push), `CaptureRreaPositionsBeforePush` (just before `OneStep`), and `AfterParticlePushBeforeBoundaries` (just after).
`Source/Initialization/ WarpXInitData.cpp`	tags the startup field solve with `Initialization` purpose.
`Source/Diagnostics/FlushFormats/ FlushFormatCheckpoint.cpp` / `Source/Diagnostics/WarpXIO.cpp`	write/read the extra RREA checkpoint payload alongside WarpX's own checkpoint.

4.2 The ordering contract

The physics is only correct if the hooks run in a specific order inside one step (Figure 3): *inject seeds → snapshot particle state → stock WarpX step → RREA trajectory reconstruction and air transport → WarpX boundary handling → RREA field solve*. The preparation script literally asserts this ordering when patching `WarpXEvolve.cpp`: if a future WarpX version reorders its loop, source preparation fails loudly instead of silently producing wrong physics.

The order is a physics contract, not a convenient insertion point. RREA must see the pushed chord *while an escaping parent still exists*: a particle that leaves the domain this step still traveled its chord through air inside it, and WarpX's boundary handling would delete it first. At the other end, generic resampling must not run before the atomic secondary groups have been committed, which is why the asserted anchors include WarpX's own `doResampling`, a stage that runs inside the protected window and is not drawn in Figure 3. This "fail-closed" habit (abort rather than improvise) recurs everywhere in the project (Section 13).

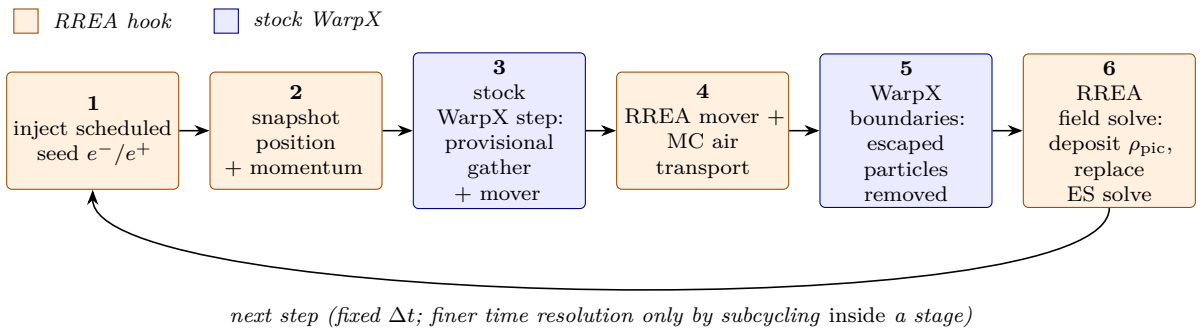


Figure 3: The per-step ordering contract. Orange stages are RREA hooks; blue stages are stock WarpX. The source-preparation script asserts this sequence (plus WarpX's own `doResampling` between stages 4 and 5, not drawn) when patching `WarpXEvolve.cpp`, and aborts if a WarpX update reorders its loop. The numbers match the walkthrough in Section 6.

5 Species and the Input Deck

The model manages three WarpX species, declared in a completely ordinary WarpX input deck (template: `cases/rrea/production/inputs`):

```
particles.species_names = rrea_electrons rrea_photons rrea_positrons
rrea_electrons.species_type = electron
rrea_photons.species_type   = photon
rrea_positrons.species_type = positron
```

All three use `injection_style = none`: WarpX never injects these particles itself. The RREA code owns creation and physical terminal decisions; WarpX's boundary pass removes escaped slots afterward. At startup the coupling verifies that each named species really has the mass/charge it expects and that the three names are distinct: another fail-closed check.

Everything else the model needs arrives as `rrea.*` input parameters. The important groups:

Group	Representative parameters
Ownership	`rrea.enable=1`; the field is the coupled Maxwell RZ Yee solve, and external particle fields are forbidden: the field may come only from the RREA solver.
Transport tables	`rrea.interaction_table_config` (path; the configuration names its own transport model); photon feedback and positron transport are unconditional and have no deck keys, and the photon cutoff and hard-Møller threshold are the transport configuration's own. Optical-depth guards (`rrea.transport_max_optical_depth_per_substep`, `rrea.transport_max_subcycles`; exceeding the cap aborts).
Population control	`rrea.population_ceiling_policy=adaptive_resample_v1` (mechanism and the scope of its unbiasedness in §13) with `rrea.population_target_electron_macros`, `rrea.max_electron_macros`, `rrea.max_photon_macros`, `rrea.max_positron_macros` as fail-closed backstops above the control band; `strict_abort_v1` is the engine default and the policy of an explicit reference run; `rrea.low_energy_cutoff_eV` overrides the bundle's charged floor for a study (§11).
Field / Maxwell	`rrea.acceleration_field_v_per_m`, `rrea.acceleration_z_max_m` (imposed profile), mesh spacing and timestep.
Atmosphere	`rrea.air_density_model` (a constant density ratio at a chosen altitude, or a full altitude profile), the density ratio ρ/ρ_0 itself. The air composition is fixed by the tables' G4_AIR reference, not by deck fractions. See Section 7.1 for how one table set serves every altitude.
Seeding	`rrea.seed_schedule_path` (a CSV of when/where/how many seed electrons and positrons appear, at what energy, weight, and direction; the optional `species_id` column routes each event to its container), `rrea.rng_seed`. The production default schedule is the PARMA cosmic-ray e^-/e^+ source with its $t = 0$ stationary bath (§15, Deposition and population). `parma_plus_1mev_pulse` adds one 1-MeV electron pulse to that same background; PARMA continues throughout the run, including after quenching. The C&D cohort is separate, without PARMA.
Diagnostics	`rrea.diag_interval`, `rrea.output_dir`, `rrea.video_diag_*` (Section 9).

The *macroparticle weight* deserves a beginner's note: one computational particle represents w real particles (w is set by the seed schedule; secondaries inherit the parent scale). Larger w means fewer

macros (cheaper, noisier), smaller w the reverse.

One more setting a newcomer should know about: the *time step* is fixed for the whole run (`warpx.const_dt`). There is no adaptive global stepping: wherever finer time resolution is needed, the relevant layer subcycles internally (the Monte Carlo transport under its optical-depth guard; the coupled fluid/Maxwell advance under its conductivity, diffusion and drift limits), keeping the outer step sequence shared by every coupled subsystem and checkpoint.

6 One Time Step, Start to Finish

Figure 4 shows the whole per-step data flow at a glance; the numbered walkthrough below follows the same arrows in execution order.

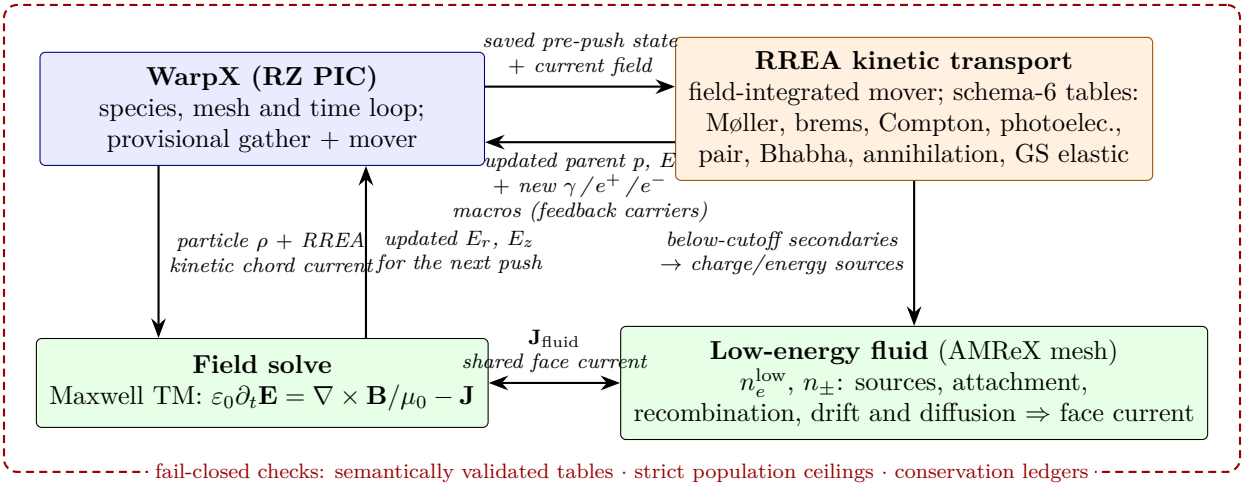


Figure 4: Per-step data flow. WarpX executes its stock step provisionally; the RREA engine reconstructs the physical trajectories from the saved state, applies air physics, and overwrites the endpoints, promoting above-cutoff secondaries to new macroparticles (which closes Dwyer's relativistic feedback loop) and demoting below-cutoff secondaries into the low-energy fluid; the fluid and kinetic chord currents drive the Maxwell TM update, whose field feeds the next particle push. The dashed frame is the fail-closed checks around the coupled path (Section 13); the execution order within one step is the hook sequence of Figure 3.

All coupling state lives in one singleton, `rrea::warpx::RreaWarpXCoupling`, declared in `native_amrex_rrea/warpx_overlay/Source/Rrea/RreaWarpXCoupling.H`. Its members are defined across sibling translation units, one responsibility each: the field step (`RreaWarpXCoupling.cpp`), startup contracts and table/schedule loading (`RreaCouplingConfig.cpp`), population control (`RreaCouplingPopulation.cpp`), seed and event sources (`RreaCouplingSources.cpp`), the reduced CSV (`RreaCouplingDiagnostics.cpp`), video frames (`RreaCouplingVideoFrame.cpp`), the C&D plane stream (`RreaCouplingPlaneWriter.cpp`), checkpoint/rank-state I/O (`RreaCouplingCheckpoint.cpp`) and the continuity fixture (`RreaActualContinuityFixture.cpp`). Shared helpers sit in `RreaCouplingDetail.H`; the one rank-local sweep over live macroparticles (`RreaForEachValidParticle`, `RreaParticleSweep.H`) gives the soft-deleted-slot guard a single owner. Its per-step entry points fire in this order:

1. `InjectScheduledSeeds`: reads the seed-schedule CSV and creates any seed macros whose injection time falls in this step, routed by the schedule's `species_id` into the `rrea_electrons` or `rrea_positrons` container.

2. `CaptureRreaPositionsBeforePush`: copies every managed particle's position *and momentum* into extra particle components (`rrea_prev_*`): the pre-push state the transport below re-integrates from.

3. **WarpX `OneStep`**: the stock WarpX gather and relativistic push still execute, but the kick is *discarded*: the engine rebuilds direction, energy and geometry from the captured pre-push state, drives its own running chord with the field impulse re-applied per substep (half-kicks around the chord, first order in the step; Gauss–Legendre line-integral work), and overwrites WarpX's endpoint with the point the particle actually reached. `algo.particle_pusher` is therefore inert plumbing; `algo.particle_shape` is *not* (the hook's own charge deposition uses it).

4. `AfterParticlePushBeforeBoundaries`: the material physics. For each managed particle, the Monte Carlo engine (`RreaParticleInteraction::Apply`) drives the running chord through air from the pre-push state. `Apply` only builds the per-step context (`TransportStep`, `RreaTransportStep.H`) and runs four phases: the electron, photon and positron loops (one shared tile loop, `TransportSpecies` in `RreaSpeciesTransport.cpp`), then the commit tail (`RreaSecondaryCommit.cpp`). Every charged particle, live or newborn, electron or positron, runs the one `AdvanceChargedHistory` and every photon the one `AdvancePhotonHistory` (`RreaKineticStep.H`); the drag, elastic and final-state physics is shared (`RreaChargedFinalStates.H`, `RreaPhotonFinalStates.H`, `RreaMaterialPath.H`, `RreaPathFieldWork.H`). Along the chord:

   - continuous energy loss (collisional + radiative drag) and field work are integrated along the chord;
   - discrete interactions compete as hazards: the traversed *optical depth* in each hard channel (Møller, bremsstrahlung, photon Compton / photoelectric / pair production, positron Bhabha / annihilation) is accumulated, and the engine samples which event, if any, occurs and where on the chord. Goudsmit–Saunderson elastic deflection is accumulated separately over each sub-interval;
   - sampling uses the **schema-6 tables** (Section 7) with log-log interpolation, local air-density scaling (Section 7.1), and *forbidden* extrapolation;
   - secondaries above the cutoffs become new macroparticles of the appropriate species; charged secondaries *below* the cutoff are "demoted": their charge and energy are deposited into mesh accumulators as low-energy-electron and ion sources (cutoffs: §11);
   - every state-changing discrete interaction closes a charge and energy ledger; a violation aborts;
   - randomness comes from a counter-based RNG keyed on the physical identity of the particle. Existing identities retain their streams, but newborn ID assignment can change with traversal order and MPI layout;
   - before any group of new particles is committed, all ranks plan exact-CIC control over survivors plus staged charged candidates, then collectively preflight both charged plans and all population ceilings (Section 13).

5. **WarpX boundary handling**: stock WarpX removes particles that left the domain.

6. `BeforeFieldSolve`: called where WarpX would solve its electrostatic field; returns `true`, so WarpX's solver never runs. Instead:

   - the hook deposits the energetic particles' charge itself: in the electrostatic branch stock WarpX pushes with deposition skipped, because the stock design deposits inside the very solver this hook replaces;
   - the accumulated event sources (low-energy electrons and positive ions) are normalized and handed to the fluid;
   - `RreaAmrexAdvance::AdvanceOneStep` evolves the fluid and the field (Section 8);
   - the resulting E_r, E_z are copied back into WarpX's field MultiFabs (so the next step's particle push feels the updated field);
   - video frames and the reduced diagnostics row are written if due.

7 The Schema-6 Interaction Tables

Process rates and several energy/final-state distributions come from pre-generated data compared with Geant4 10.7.4 and organized under `rrea_transport_tables/`. Runtime analytic models remain explicit where used (notably free Klein–Nishina Compton and bounded Modified–Tsai bremsstrahlung angles). The table directory is selected through `transport_physics.json` and the loader checks ranges and cutoffs; focused tests compare the physical holdouts.

The shared electron/positron brems angle uses the Modified–Tsai distribution of `G4ModifiedTsai` in the pinned Geant4 source: its Gamma-mixture variable u is conditioned on $0 \le u \le 2\gamma$, then mapped by $\cos\theta = 1 - u^2/(2\gamma^2)$, where $\gamma = 1 + K/(m_e c^2)$. This analytic angle is independent of the tabulated photon energy; it does not claim option-4's energy–angle correlation. It requires no angular table.

- **Electrons**: restricted collisional drag, soft-radiative drag, hard Møller mean free path + secondary spectrum, bremsstrahlung MFP + photon energy CDF + analytic emission angles, Goudsmit–Saunderson elastic multiple-scattering angle CDFs;
- **Photons**: modeled-channel interaction MFP and probabilities, photoelectric subshell CDFs (with binding energies), nuclear and triplet pair production with replayed Geant4 final-state sample banks. Compton is the one channel whose final state is *not* tabulated: its rate carries Geant4's incoherent scattering function (measured $-4.4\,\%$ against free Klein–Nishina at 50 keV), but the scattering angle is sampled at runtime from analytic free-electron Klein–Nishina;
- **Positrons**: *restricted* collision stopping (its own table: up to 15 % above the electron value below 100 keV, within 0.5 % above 1 MeV), soft-radiative drag, hard Bhabha, positron bremsstrahlung, in-flight and at-rest annihilation.

The photon and positron channels are what close the relativistic feedback loop, and the engine makes that unconditional: every *explicitly produced* bremsstrahlung photon is a tracked particle

able to Compton-scatter or pair-produce anywhere in the domain (including *behind* the avalanche front), and pair-produced positrons (which the ambient field accelerates backward) are transported with their own drag, hard-Bhabha, bremsstrahlung and annihilation physics. There are no deck keys for either (§5), so feedback is not a bolted-on approximation (no "feedback factor"): it *emerges* from the same per-particle Monte Carlo transport as the primary avalanche.

Coherent Rayleigh scattering is deliberately excluded from the schema-6 kinetic gamma-ray transport: it has no runtime channel and no rate or angular table. Near-cut photon mean free paths, angular transport, and feedback predictions are conditional on that scope. This does not describe the separate optical CloudScat calculation, whose atmospheric Rayleigh channel is enabled unless explicitly disabled.

"Every explicitly produced" is meant literally, and the qualifier matters. Bremsstrahlung photons below the tracked cut are not created as particles at all; they are carried as continuous radiative drag on the parent. That cut is $k \geq$ `brems_absolute_cutoff_eV`, a flat energy declared by the loaded transport configuration; the engine binds to it and pins no value, so quote the number from the bundle a result came from. Energy is conserved: the sub-cutoff loss is booked as `electron_soft_radiative_stopping` and is radiated away, not converted into ion pairs (a 1e-3 share of the deposit in production). Photons that Compton-degrade or annihilate below the same cut are the opposite case: their energy is deposited locally as ion pairs, a few hundred metres from where it would really be absorbed. In both cases the photon *count and spectrum* are incomplete below the cut, where the tracked population is only Compton-degraded photons. Any statement about photometer counts, low-energy irradiance, or the backscattered feedback carriers (which pile up below $m_e c^2/2 = 255\,\mathrm{keV}$) must carry this caveat.

7.1 One table set for every altitude: density scaling

The tables are computed once, for Geant4 `G4_AIR` at $\rho_0 = 1.20479\ \mathrm{kg/m^3}$ (dry air, 20 °C, 1 atm), the value the transport configuration records as `reference_density_kg_m3`. It is deliberately not called "STP": the RREA threshold and reduced ion mobilities use their own reference densities (below and Section 12), and every ratio in the code names its reference. But avalanches live at thundercloud altitudes where the air is several times thinner, so how can one table set be correct everywhere? The answer is that as long as the *composition* of the air is fixed (the N_2/O_2 ratio of the tables' G4_AIR reference), every interaction rate is simply proportional to how many molecules a particle passes per meter. The engine therefore evaluates the local density ratio ρ/ρ_0 at each particle's position and rescales on the fly: mean free paths stretch by ρ_0/ρ (thinner air = fewer collisions per meter), while drags and other inverse lengths shrink by the same factor. This rescaling is exact for every rate whose only density dependence is that molecule count. The lone exception is the Sternheimer density-effect term inside the collision stopping table, which stays frozen at its ρ_0 value: it is identically zero below ≈28 MeV (hence exactly zero at the ≈1.7 MeV friction minimum that sets the RREA threshold, so the threshold field itself rescales exactly) and reaches only ≈2.9 % by 100 MeV, with the sign such that the drag is *under*-estimated.

Where ρ/ρ_0 comes from is a deck choice (`rrea.air_density_model`): either a single constant ratio for explicit diagnostic inputs, or a full altitude profile: `RreaAltitudeProfile` in the native library interpolates a CSV of density versus altitude, with the mesh's z coordinate mapped to altitude above sea level. The canonical capture and all C&D comparisons use the realistic altitude profile, never a constant-density substitute. The same profile machinery can shape the imposed background

field. The low-energy fluid follows suit: ion mobility is scaled by the same local density, re-expressed against the Loschmidt density ρ_L at which reduced mobilities are quoted, and the electron mobility and attachment are read from the source-anchored E/N closure table, whose reduced field carries that density, so the chemistry and conductivity are consistent with the transport.

7.2 Loading and integrity checks

The loader (`rrea::RreaInteractionTables`, native library) validates the tables semantically before transport mutates state: required files and columns, ranges, monotone energy/CDF grids, finite positive rates, normalized probabilities, cutoff agreement, and model identity. Any mismatch aborts the run. Off-table energies are an error, not a clamp. Schema-6 contains *no* calibrated fudge factors. After validation `BuildKineticStorage()` binds every table to a fixed `RreaTable` enum slot, so the hot path indexes an array rather than looking a name up.

7.3 Geant4-derived transport data and validation

A natural beginner's question is why Geant4 is not called inside WarpX. Geant4 is a full transport framework with its own geometry, stepping, and threading model; embedding it in a PIC loop would be impractical. The project therefore samples a versioned schema-6 table set produced from a declared Geant4 reference. Geant4 itself never executes inside WarpX.

The only production data tree is `rrea_transport_tables/schema6/production/`, tracked in the repository as RREATBL containers (`native_amrex_rrea/include/rrea/RreaTableContainer.H`): each file is a JSON header naming every column with its dtype, over one deflate stream of byte-shuffled float64 columns. `scripts/rrea_table_pack.py` writes them and dumps any column back to CSV for review. Nothing is narrowed: a float32 store would collapse the sub-ulp energy anchors the loader requires to stay distinct. `transport_physics.json` stays plain text: it carries model identifiers, cutoffs, acceptance tolerances and the path of every table, which is where a replacement set is named.

At run time `RreaInteractionTables` loads the validated tables: a rate such as "mean free path of a 3.7 MeV electron against hard Møller scattering" is interpolated log-log between grid nodes, and when an interaction fires its final state is drawn by inverting the appropriate CDF.

8 The Low-Energy Fluid and the Field Solve

8.1 Fluid state

`rrea::LowEnergyFluidState` (native library) holds, per mesh cell: low-energy electron density n_e^{low}, positive and negative ion densities $n_\pm$, their source rates (fed by the Monte Carlo demotions and ionization), the conductivity σ (whose local Maxwell relaxation time is $\tau_m = \varepsilon_0/\sigma$), and the fluid charge density. Each adaptive material substep uses a symmetric half-reaction/transport–Maxwell/half-reaction split. The outer-step source-rate fields remain intact until the final half reaction, so their known integral is distributed exactly across the substeps; table attachment is re-evaluated after each updated field. The transport stage computes $\sigma = e \sum_s \mu_s n_s$ with mobilities

scaled by air density, summed only over carriers that are actually advanced (a species counted in σ but held stationary would break charge continuity); and drift the carriers with adaptive shared, r-weighted finite-volume substeps using the committed donor-upwind face currents that enter the corresponding Maxwell substeps. The code aborts on any substep whose multidimensional outgoing carrier CFL exceeds the configured limit after positivity limiting.

8.2 Why the field solve is different from stock WarpX

WarpX's electrostatic solver answers "what field does this charge make?" The thundercloud problem needs more: an *imposed* background field (the cloud's charging, held fixed as a boundary/reference state) that the plasma progressively screens. The RREA solver therefore replaces that electrostatic solve entirely with an explicit finite-speed update: the ambient field is built once as a curl-free background (Section 8.4) and the total field is then advanced by the RZ TM Maxwell equations (Section 8.3).

Screening is a consequence of that advance rather than a term in an operator. Where the air has become conductive, its current enters Ampère's law as $-\mathbf{J}/\varepsilon_0$, relaxing the field on the Maxwell time $\tau_m = \varepsilon_0/\sigma$ because Ampère's law says it must. Material continuity is a fail-closed outer-step gate and Gauss a monitor; neither applies a correction.

8.3 The Maxwell TM branch

Every current here is poloidal, so under $m = 0$ the complete retained field content is the TM triplet (E_r, E_z, B_θ), advanced on a staggered Yee mesh by explicit leapfrog: E_r on radial faces, E_z on axial faces, B_θ on corners, so both curls land exactly where they are consumed with no averaging. The solver advances only the *scattered* field over the static thundercloud ambient; their sum is what chemistry, drift and the particle gather see, the fluid consuming the exact face fields, while the particle gather reads a cell-centered two-point face average (one extra half-cell-symmetric smoothing relative to the faces Ampère evolves). The source is $\mathbf{J} = \mathbf{J}_{\text{fluid}} + \mathbf{J}_{\text{kinetic}}$, where $\mathbf{J}_{\text{fluid}}$ is the fluid's own committed donor-upwind face current (the same object that moves the carriers), so Ampère's law and continuity cannot disagree about how much charge moved, and there is no separate σ term to double-count. The explicit coupling reports the maximum over its internal substeps of `em_max_dt_sigma_face_over_eps0`, the explicit diffusion number, and the reaction spectral number. Targets, hard guards and the allowed substep range are resolved from the run configuration. The count is recomputed from all three stiffness estimates each outer step. The initial probe uses $n_e + \Delta t S_e$ and $n_+ + \Delta t S_+$ in existing transport scratch, anticipating ionization without committing carriers or changing chemistry; mobilities are frozen for that estimate and realized substep guards remain. Particle, video and checkpoint timesteps are unchanged. The energy ledger lives on the one discrete RZ metric of the staggered mesh (dual volumes, `RreaMaxwellTMRZ.H`): Joule work and the boundary term are bilinear means over each Ampère substep, the boundary term being the exact discrete remainder of the summation by parts (face-aligned Poynting products plus the Faraday work of the absorber-owned corners), so that in the leapfrog's mixed-time form $\Delta U_E + \Delta U_B + \Delta t\,(P_{\text{Joule}} + P_{\text{out}}) = 0$ closes to round-off; the native Maxwell smoke pins that identity and exact closed-domain conservation. The reported ledger keeps B_θ^2 at the outer half-time, so its residual is the $O(\Delta t^2)$ magnetic-stagger term. The kinetic chord current is constant over the outer step. B_θ is re-staggered into the internal spacing and back to the outer $t - \Delta t/2$ convention before checkpoint/radio output. Boundaries are first-order Silver–Müller; the native Maxwell smoke

measures their residual field energy after an outgoing pulse. The axis is regular by construction ($E_r(0) = B_\theta(0) = 0$, and E_z is never sampled on the axis).

Material charge continuity is enforced before the outer step is committed: the actual WarpX particle deposit plus persisted fluid charge at both time levels must close cell by cell against the time average of the exact substep face currents handed to Ampère's law. The mixed norm is 10^{-10} relative with a binary64 cancellation floor of $8\sqrt{\epsilon_{64}}$ times the heaviest live macro charge, swept from the population each step: that quantum sets the scale of the resampling residue, and losing a whole macro would still be $\sim 10^7$ floors. Only after the check passes may the outer step be published. Seeds are injected without a current, so the field is that of neutral pairs created in place (right for secondaries produced locally in the air); the separate Gauss monitor measures the counter-charge they leave behind, not a field error.

8.4 The imposed background and its lateral extent (radial taper)

The imposed field enters as a background potential ϕ_{bg} (`InitializePhi`, `RreaFieldInitializer.H`), from which *both* ambient face families are realized as differences of that one potential (`InitializeBackgroundState`, `RreaAmrexAdvance.cpp`). Because both come from the same ϕ_{bg}, the discrete $(\nabla \times E)_\theta$ telescopes to exact cancellation, so the ambient is curl-free by construction rather than by assertion, and the code verifies it at every interior B_θ corner, failing closed at initialization if it ever stops holding.

The background supports a **radial taper** (`rrea.field_taper_r_start_m` / `rrea.field_taper_r_end_m`):

$$\phi_{\mathrm{bg}}(r,z) = \phi_{\mathrm{ref}} + f(r)\,[\phi_z(z) - \phi_{\mathrm{ref}}], \qquad E_z = f(r)\,E_z^{\mathrm{prof}}(z), \quad E_r = -f'(r)\,[\phi_z(z) - \phi_{\mathrm{ref}}],$$

with f a C^2 quintic smoothstep (f=1 for $r \le r_{\mathrm{start}}$, f=0 for $r \ge r_{\mathrm{end}}$) and ϕ_{ref} the midrange of ϕ_z over the domain (minimizing and symmetrizing the fringe E_r). Because the taper is applied at the *potential* level, the background stays curl-free, and the implied ρ_{ext} is a physical charge configuration: the profile's charge layers laterally confined to $r < r_{\mathrm{end}}$ plus the annulus fringe charge: a storm charge region of finite lateral extent instead of infinite slabs terminated by a charged wall. The wall column (f=0) is pinned at the constant ϕ_{ref}: a fixed ambient potential at the radial boundary; the scattered field obeys the absorbing boundary condition. In the taper annulus the ambient field can drop below the local density-scaled RREA threshold, allowing the avalanche front to stall before it reaches the wall.

The fade width is not a free choice: the circulation constraint. The fringe field is not an implementation detail: for *any* curl-free field that carries the full profile at small r and vanishes at large r, $\oint \mathbf{E} \cdot d\boldsymbol{\ell} = 0$ forces the entire column potential across the fade annulus, so

$$\max |E_r| \;\simeq\; 1.875\,\frac{(\max \phi_z - \min \phi_z)/2}{r_{\mathrm{end}} - r_{\mathrm{start}}}$$

(the 1.875 is the quintic smoothstep's peak slope; changing the taper shape does not remove the circulation constraint). A usable fade width therefore satisfies $w \gtrsim 1.875\,\Delta\phi_z/(2\,E_{\mathrm{acceptable}})$. The driver audits the fringe against the local density-scaled runaway threshold, aborting on any super-threshold pocket. The loaded vertical profile is already approximately zero at the domain floor and exactly zero above its upper support, so it needs no separate vertical taper.

8.5 Residual ion drift and delayed field enhancement: the E125 pulse

The first 500 μs of OLIVIA run `runs/e125_pulse` show that ion motion remains electrically important after the first gamma-ray pulse.

The snapshot below (recorded source `53fecbd`) has a single initial pulse of 10^{12} electrons at 1 MeV, an initial peak field of 125 kV/m, the altitude-dependent atmosphere, and ion drift enabled. Its domain spans 6–16 km altitude and $r < 10$ km, with $\Delta r = 5$ m, $\Delta z = 6.94$ m and $\Delta t = 2.5$ ns. The assessment uses `command.json`, `rrea_reduced.csv`, the saved video profiles, `rrea_video/photon_band_series.csv`, and the fluid/Maxwell state saved at 500 μs. The optical assessment below includes the later millisecond continuation; the estimates in this subsection concern the earlier snapshot.

The ion fluids have drift velocities $\mathbf{v}_\pm = \pm\mu_\pm \mathbf{E}$, with $\mu_\pm = K_\pm / \max(\rho/\rho_L, 0.05)$, $K_+ = 1.4 \times 10^{-4}$ and $K_- = 1.9 \times 10^{-4}$ m^2/(V s), where ρ_L is the Loschmidt reference density. Opposite ion velocities produce conventional currents in the same direction:

$$\mathbf{J}_{\text{ion}} = \sigma_{\text{ion}}\mathbf{E}, \qquad \sigma_{\text{ion}} = e(\mu_+ n_+ + \mu_- n_-), \qquad \tau_{\text{ion}} = \frac{\varepsilon_0}{\sigma_{\text{ion}}}.$$

The local positive-ion loss time from ion–ion recombination is $\tau_{\text{ii}} = 1/(\alpha_{\text{ii}} n_-)$, with this run's $\alpha_{\text{ii}} = 2 \times 10^{-12}$ m^3/s. These are local characteristic times, not a clock predicting a second pulse. At 500 μs, the first radial cell ($r = 2.5$ m) gives the following rounded estimates; positive v_z is upward. The first row is the enhanced-field front and the second the densest residual ion region.

Altitude (km MSL)	$\lvert E\rvert$ (kV/m)	$v_{+,z}$ (m/s)	$v_{-,z}$ (m/s)	τ_{ion} (ms)	τ_{ii} (ms)
11.66	192	−99	+134	2.5	27
12.64	7.2	−4.2	+5.7	0.050	0.62

These velocities are inferred from the saved field and mobility law, not individual-ion trajectories. At the front they would give only 0.05–0.07 m of travel in 500 μs, or 0.30–0.40 m in 3 ms, *if the field persisted*; motion in the screened region is much slower. Nevertheless, that dense region retains $n_+ \simeq n_- \simeq 8.1 \times 10^{14}$ m^{-3}, and ions supply 99.9% of its conductivity. Small charge separations can therefore change the field well before recombination removes most carriers. A moving ion-density centroid or discharge front is not a measurement of individual-ion travel: ion creation and spatially varying losses also move those features.

Field redistribution is already present. Near 11.66 km the field grows from about 120 to 192 kV/m, while near 12.64 km it falls from about 105 to 7.2 kV/m. Screening behind an enhanced-field front is the physical meaning of "field compression" here. However, the field maximum levels off near 193 kV/m by 300–400 μs; this record does not show uninterrupted rapid amplification. Video endpoint quantities refer to the nearest saved frame, at 499.92 μs.

The photon tail recovers weakly. The instantaneous population of photons above 1 MeV in the 12–16 km band, integrated over $r < 10$ km, peaks at 2.43×10^{17} near 69.5 μs, falls to 1.77×10^{12} near 364 μs, and recovers to 1.68×10^{14} at the last frame. That is about 95 times the late minimum but only 0.069% of the first peak. The recovery also appears in interval averages: the mean population rises from 5.52×10^{12} over 250–300 μs to 9.27×10^{13} over 450–500 μs. This observable is a population

inside a volume, not an escaping flux or detector light curve. The record supports late renewed activity, but neither a second large pulse nor attribution solely to ions.

Relation to Dwyer's mechanism. Dwyer (2005), *The initiation of lightning by runaway air breakdown*, paragraphs 6–8 and 14, couples ion polarization to fresh runaway ionization and positron feedback; its advancing discharge front can move much faster than the ions. The reported 3.3 ms is the rapid field-amplification interval after storm charging, not a predicted recurrence period between gamma-ray pulses. That example reaches about 1200 kV/m sea-level equivalent. The E125 snapshot reaches about 708 kV/m equivalent, using $E_{\mathrm{eq}} = |E|(1.293\ \mathrm{kg\,m^{-3}})/\rho$. The initial single-pulse source differs from Dwyer's sustained cosmic-ray seeding, and this fluid omits ion collection by cloud droplets and subsequent conventional discharge development.

Scientific conclusion. Continued ion-driven field redistribution and a later gamma-ray pulse are physically plausible, but their occurrence, amplitude and delay are unestablished by this snapshot. Later pulse-only and PARMA-plus-pulse records are used in the optical assessment below. A matched run with ion drift disabled would test causal attribution; spatial, timestep and macroparticle convergence are still needed for quantitative claims. Neither a millisecond continuation nor that control is part of the evidence above.

9 Diagnostics, Video, and Checkpoints

Reduced CSV (`rrea_reduced.csv`) One row every `rrea.diag_interval` steps with core volume-integrated quantities and configuration-dependent extras: species populations and physical weights, fluid totals, peak conductivity, field-fraction statistics (the `field_fraction_*_accel_region` columns span the profile-active band inside the taper core; a channel-local field needs the frame stream), per-channel interaction counters, and the energy/charge ledger totals, among them `em_charge_continuity_signed_accum_e`, the running signed continuity residue. Global-population, optical-source and gate diagnostics read this file.

Video diagnostic When enabled, the coupling resamples selected quantities onto a fixed $n_r \times n_z$ display grid at scheduled simulation times. The `rrea_video_frames_binary_v4` contract stores exactly five arrays, in order: energetic-electron density above the threshold, positive-ion density, $|E|$, photon density above the threshold, and positron density above the threshold. Frames append to `video_frames.bin` with CSV/JSON sidecars, whose metadata lists the arrays in order. These are display resamples, not state, so v4 narrows them to `float32` and byte-shuffles and deflates each frame independently (`RreaVideoFrameCodec.H`, decoded by `scripts/rrea_video_stream.py`). Compression is data-dependent, and the only precision loss is the declared `float64`-to-`float32` display conversion. Frames are therefore variable length, and the `byte_offset`/`byte_length` columns of the per-frame CSV are the index, which keeps the restart trim an exact truncation at a frame boundary, the property the cluster/laptop handoff depends on. Rendering to MP4 is a separate offline step (`scripts/render_rrea_profiled_video.py`); capture resolution is fixed at run time, but colormaps and styling are render-time choices. The standard video's left profile axis uses one fixed scale for the initial radial mean (gray dashed), current radial mean (blue), and current first-cell "$r = 0$" slice (red).

Checkpoint v2 Alongside each WarpX checkpoint, the coupling writes an `rrea_checkpoint/` payload: all fluid MultiFabs, the previous PIC charge state, the scattered Maxwell state, and

per-rank RREA counters (including signed-plane diagnostic counters and append position). The checkpoint reader owns format, dimensions and state consistency: it verifies the completion marker, the schema and format versions, and the mesh/state dimensions of every field it reads. (Simulation time and step continuity are WarpX's own; the RREA payload neither writes nor checks them.) A *rank-count* change requires refolding rank-local counters, so it aborts unless the run opts in with `rrea.allow_restart_epoch_change=1` and a reason string (what the driver's `--restart-epoch-break` emits); the counters then fold deterministically onto the new layout; if the reduced-CSV header changes across the break the engine archives the superseded stream itself (`ArchiveCsvIfHeaderChanged`, `RreaCouplingDiagnostics.cpp`). `docs/RREA_RERUN_GUIDE.md` carries the operator recipe.

10 From Simulation to Radio, Longitudinal-Field and Optical Observables

The engine outputs populations, fields, and currents. Offline analysis uses the Maxwell boundary record for direct off-axis free-space radio, an exact-axis longitudinal-field diagnostic, and the native ion-pair source for optical estimates.

10.1 EM fields from the recorded Huygens surface

A Maxwell capture appends the scattered (E_r, E_z, B_θ) on a closed surface **inset eight cells** from the absorber to `rrea_em_boundary.bin` at fresh initialization (zero scattered state) and every completed step, and `scripts/compute_rrea_em_radio.py` propagates it to an observer by a retarded surface integral (Love equivalence: $\rho_s = \varepsilon_0 \hat{n}\cdot\mathbf{E}$, $\mathbf{M}_s = -\hat{n}\times\mathbf{E}$, $\mathbf{J}_s = \hat{n}\times\mathbf{H}$). Three details are load-bearing. The surface must be inset, because on the boundary itself Silver–Müller slaves B_θ to the tangential E, so a record taken there carries no independent magnetic information. B_θ is a leapfrog half-step behind E, so it is interpolated on its own time axis $(t - \Delta t/2)$; treating the two as simultaneous shifts the magnetic record by half an outer step. And the recorded field enters the kernel raw: $\rho_s = \varepsilon_0 \hat{n}\cdot\mathbf{E}$ cancels the $1/4\pi\varepsilon_0$ of the Jefimenko charge terms and $\mathbf{H} = \mathbf{B}/\mu_0$ cancels the μ_0 of the $\partial_t \mathbf{J}$ term, so neither constant appears in the integrand. Reinstating either scales that family of terms by $\varepsilon_0^{\pm 1}$.

Version 4 timestamps label the completed electric-field step. The engine writes only version 4, and a restart drops the frames at or after its own start time before appending. The integral returns only the sources the surface encloses; the exterior-source limit is in `docs/RREA_RADIO_LIMITATIONS.md`.

The on-axis output is not radio. For an $m = 0$ source E_r and B_θ vanish on the symmetry axis, so the Poynting flux there is identically zero. The exact-axis E_z this integral returns is a longitudinal Coulomb/induction field, which is what an overhead field mill or nadir field-change meter measures, not a propagating waveform. A radio prediction needs an off-axis observer and the full transverse vector integral with azimuthal quadrature; a distant on-axis waveform alone can therefore never establish a self-quenching claim.

10.2 Radio scope

For a nonzero observer offset, `compute_rrea_em_radio.py` reconstructs the full vector Love-equivalent surface sources around each RZ ring and performs azimuthal retarded quadrature. The $m = 0$ symmetry does not collapse that integral off axis: it makes the field values ϕ-independent, while the observer distance and hence the retarded time are not. The series is truncated to the window every quadrature node can answer from the record, because extrapolating past the last recorded frame would invent a boundary source and an artificial pulse tail. It writes $\mathbf{E}$, $\mathbf{B}$ and $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ in free space. No ground image, ionosphere, terrain or antenna gain is applied. A geometric projection onto the ALOFT EFCM (the ER-2's flat-plate electric-field-change meter) and nominal unit-gain fast/slow responses are instrument scenarios, not calibration. The evaluator stages the required chronological source brackets in one reused contiguous RAM slab per observer, retaining the global E/B time origins. The timestamp index is read once in bounded sequential chunks, avoiding strided network-file reads. Both observer paths use `retarded_surface_sum` (`scripts/rrea_retarded_sum.py`): Numba compiles the same piecewise-linear interpolation and derivative kernels, parallel over observer times, without fast-math reassociation. Geometry, azimuthal quadrature and output cadence are unchanged. The slab spans 4096 output samples plus their delay envelope, not the whole capture. The optional azimuth check compares N with $2N$ for both E and B. This IS the source of the multiband figure's waveform and spectrum panels: a host-side reduction (`cluster/olivia/slurm_rrea_em_observers.sbatch`) writes one series per observer (a 100 km ground receiver, the ER-2 at 20 km MSL, and the nadir field mill on the exact-axis path) beside the band series, and only those small CSVs travel. It runs where the multi-GB record lives and is kept out of the render job. The series carry OBSERVER time; the figure removes the geometric delay to put them on its source-time axis. The compact-source current-moment waveform stays as the dashed reference on ground bands below the radial corner, and carries the panels outright for a capture whose reduction has not been run, which every panel title names. Both paths see a *truncated* source: an electron absorbed at the top boundary stops contributing to $M(t)$, so an escape burst injects a spurious $\mathrm{d}M/\mathrm{d}t$. The escaped-charge diagnostic flags truncation but does not bound its waveform error.

Video diagnostics. `scripts/extract_rrea_band_series.py` reduces the frame stream on the capture host to band population series for the video and the multiband figure: both species everywhere above 12 km MSL over the full mesh radius. A whole-population observable needs a large volume, not a sample at one altitude, because a thin band sits in the macroparticle-occupancy tail and is shot-noise limited there. Each is an instantaneous population, not a crossing flux; a leading zero interval is a sampling-occupancy floor. The render job refreshes the sidecar before every render and drops a stale or missing strip rather than drawing it. The renderer also draws the optical source or irradiance series described below, but no radio waveform.

For a creation map, `rrea.electron_creation_bin_m=25` enables a 25-m-bin weighted first-passage diagnostic at 1 MeV. Each electron contributes once, at an energetic birth or its first upward crossing; numerical splits inherit the flag. Crossings combine field work and drag, excluding intermediate half-kick overshoots. Flags and cumulative grids survive restart; each captured frame also saves a lossless snapshot in `electron_creation_history.bin.gz`. `scripts/plot_rrea_electron_creation.py` plots the full accumulation and interval densities $[N(t_b) - N(t_a)]/\Delta V$ on a shared scale, using exact annular volumes $\Delta V = \pi(r_{\mathrm{hi}}^2 - r_{\mathrm{lo}}^2)\Delta z$. Intervals are $(t_a, t_b]$, with $t_a = 0$ also including initial injections.

Automatic bounds divide first-passage weight approximately equally; `--bounds-us 0,20,50,100,300,1000` snaps explicit bounds to saved frames. Counts include energetic injections and remain weighted Monte Carlo estimates, not occupancies or rates. Unequal interval durations must be considered when comparing activity. For a running capture, copy only the history prefix ending at `history_bytes` in a COMPLETE checkpoint's `electron_creation.bin`; the last plotted frame may precede that checkpoint by one capture interval.

10.3 Optical emissions at 337.1 nm and 777.4 nm

`scripts/compute_rrea_optical_emissions.py` estimates the two ASIM/MMIA photometer bands from a profiled production capture. It predicts deposition fluorescence and separately assesses omitted field-driven molecular excitation. The latter is not a multiplier on deposited energy or a second plotted prediction.

Degradation term (baseline). Every ionizing cascade converts a fixed fraction of the deposited energy into fluorescence, measured without an applied field; the working baseline is the air-fluorescence yield of the cosmic-ray literature. For the 337.1 nm band (N_2 2P(0,0), $C^3\Pi_u \to B^3\Pi_g$) the world average is 7.04 ± 0.24 photons/MeV at 800 hPa, 293 K (Rosado et al., arXiv:1103.2022; the final AIRFLY value, 5.61 ± 0.22 at 1013 hPa, 293 K, Ave et al. 2013, agrees with it through the quench law to 0.5 % and is pinned in `test_optical_physics.py`), with collisional quenching $\mathrm{FY}(P,T) = Y_0/(1 + P/P'(T))$, $P'(293\,\mathrm{K}) = 15.9\,\mathrm{hPa}$ and $P' \propto T^{0.86}$. That exponent is $1/2 - \alpha$ for a collisional cross section $\sigma \propto T^{\alpha}$, with $\alpha_{337.1} = -0.36 \pm 0.08$ measured by AIRFLY (Ave et al., *Nucl. Instrum. Meth. A* **597**, 50, 2008, Table 1); the $\alpha = 0$ ($\sqrt{T}$) convention that paper refutes overestimates the yield at altitude by ∼8 % near 9 km. Water-vapour quenching ($P'_{\mathrm{H_2O}} = 1.28\,\mathrm{hPa}$) is not modelled; this is a dry-air yield. The optical production rate and its source-weighted altitude come from the engine's native cumulative ion-pair source (`ion_pair_source_weight_cumulative` and `ion_pair_source_z_weighted_m_cumulative`); the script exits if a capture predates them. Multiplying that rate by $W_{\mathrm{air}} = 33.97\,\mathrm{eV}$ gives the deposition rate. The script prints `transport_energy_loss_eV` and its implied energy per pair as a consistency diagnostic, not a second deposition estimate: tracked-secondary energy is booked when created and again where it deposits, while bremsstrahlung never enters the ion-pair tally. Soft bremsstrahlung below the cut is therefore also absent from the fluorescence driver; its share is printed alongside.

Swarm excitation envelope. MCAIR is a separate experimental Python/Numba electron Monte Carlo used for the diagnostic comparisons below; those use 80:20 N_2/O_2 gas with O_2-assisted attachment only. Offline BOLSIG+ 07/2024 (Hagelaar and Pitchford, PSST 14, 722, 2005) uses the 46 selected Phelps $N_2/O_2/Ar$ records in `rrea_fluid_closure/phelps_swarm.txt`; the existing closure generator writes `dry_air_swarm_hybrid_v3/n2_c_excitation.csv`. The native process-rate output supplies k_C [m^3/s], not an integral of the truncated EEDF export. The source's C-state normalization is unresolved: Phelps's posted notes recommend 0.67 for the older Report 26. The export matches his posted table; applying that factor again remains a sensitivity. At 14 eV, multiplying the raw C cross section by $A_{00}\tau_C$ gives 3.16×10^{-21} m^2; also applying 0.67×0.545 gives 1.15×10^{-21} m^2, near Zubek's measured 337-nm value 1.13×10^{-21} m^2 (J. Phys. B 27, 573, 1994). This single-energy agreement does not justify an energy-independent normalization or vibrational fraction: Malone et al. (Phys. Rev. A 79, 032705, 2009) resolve a varying $v' = 0$ fraction. Thus $f_{v'=0} = 0.545$ and 1 remain sensitivities, not a measured interval; the former scales the reported

envelope by 0.545. Argon is retained because its 0.93% abundance changes k_C by 12–23% at 16–30 Td. The two-term approximation is a separate model uncertainty: tail anisotropy $|f_1/f_0|$ reaches about 0.7 at 28 Td, and calculated drift exceeds the measured closure by about 2.1%, 3.2% and 5.6% at 16, 20 and 28 Td. Numerical convergence therefore does not establish percent-level physical accuracy of k_C. The significance assessment also uses conservative vibrational fraction, normalization and population/field weighting. The Python spectroscopy owner supplies $b_{337} = A_{00}\tau_C/(1 + P/P'(T))$ with $A_{00} = 1.31 \times 10^7$ s^{-1} and $\tau_C = 37.1$ ns (Gilmore et al., J. Phys. Chem. Ref. Data 21, 1005, 1992). The sampled model envelope is $Q_{\rm sw} \le N_e \max_{E/N,N,T,\rm model}[x_{\rm N2} N k_C b_{337}]$, using every field knot through the next enclosing node, including interior maxima. Quenching of the deposition population is bounded independently. Convex and concave envelopes of the altitude-dependent fluorescence yield use the native ionization-source first moment and the recorded domain; positive-ion density differences are never used as source estimates.

The numerical tail is included as a conditional decreasing-tail envelope, not set to zero when the solver truncates below threshold. Grid refinement and truncation precision must converge excitation itself. Missing assessment columns or unsupported gas/field conditions produce *swarm unassessed*, while supported deposition fluorescence remains available. Intervals without deposition carry absolute swarm photons, not a ratio. The 1% significance target concerns this sampled molecular model; it is not an accuracy claim for cross sections, temporal sampling or cloud propagation.

Full-capture molecular assessment (27 October 2026). The available records start at 0.125 μs. E120 pulse reaches 500 μs, E125 pulse 4051 μs, and E125 PARMA-plus-pulse 2732.5 μs; the last is a snapshot of a continuing run. Their sampled realized maxima are respectively 19.625, 28.124 and 27.178 Td. The integrated sampled swarm envelopes are 0.035%, 0.190% and 0.091% of the independently bounded deposition fluence. E120's entire 0.035% is the numerical-tail allowance; the E125 and PARMA envelopes also include that allowance, below 0.035% in each capture. Refining 800 to 1600 energy points changes k_C by at most 0.92% at 16 Td and 0.56% at 20–60 Td; changing precision from 10^{-14} to 10^{-15} changes these rates by less than 0.08%. The manual prohibits sub-machine-precision settings. Fourfold temporal coarsening changes the largest integrated envelope by 0.34% of itself.

The relaxed-swarm assessment alone does not warrant a passive native optical accumulator. It does not bound field-induced changes to the prompt degradation spectrum: the measured deposition yield is a zero-field baseline. Intervals with a swarm/deposition envelope above 1% contribute only 5.6×10^{12} swarm photons in E125, versus 3.85×10^{21} modeled deposition photons; their largest swarm rate is below 4.4×10^{-8} of peak deposition. Zero-deposition intervals are reported separately: at most 1.8×10^6 photons for E120 and 6.0×10^7 for E125, within the same conditional model envelope. The C-state quenched lifetime is approximately 1–4 ns over this column, short compared with the 125-ns reduced-CSV spacing and 1-μs optical smoothing.

A separate matched MCAIR prompt-cascade stress test injects 128 field-aligned 10-keV electrons per realization into $N = 8.8 \times 10^{24}$ m^{-3}, 237-K gas, at zero field and 28 Td. Sixteen independent paired realizations, using the Zubek/Malone 337-nm emission response on the time-integrated spectra, give a 2.4 ± 1.1% yield increase per ion pair (95% interval). Less than 0.084% of the excitation integral weighted by the 337-nm emission response lies in the last 0.5–1 ns window; excited-state decay is not evolved in this diagnostic. The zero-field yield is 0.0120 unquenched photons per pair, consistent with the measured baseline within its uncertainty. This closes neither the energy-angle distribution of native condensed-loss secondaries nor the hybrid model's near-threshold ionization uncertainty. No

field-dependent fluorescence correction is adopted from this stress test. The relaxed-swarm envelope therefore does not certify the total optical prediction to 1%. The source-band pair-time-weighted mean fields in the three captures are 11.49, 10.89 and 10.91 Td; these regional statistics are indicative, not a substitute for a local prompt-response model.

The 1-μs source smoothing changes integrated fluorescence by less than 7×10^{-6} relative and shifts the integrated source centroid by less than 0.07 m in these three records.

The source-width uncertainty is larger: the native first moment bounds 337-nm deposition totals to $[1.57, 1.91] \times 10^{21}$ photons for E120 and $[3.83, 4.67] \times 10^{21}$ for E125. These bounds do not reconstruct source positions. In the recorded 0–10 km-radius, 6–16 km-MSL domain, direct travel times to an overhead 20-km observer span 13.3–57.4 μs. Centroid propagation cannot claim percent-level pulse-shape accuracy, and multiple cloud scattering adds an unbounded delay tail. Positive-ion profiles cannot retrospectively supply the missing emission distribution.

777.4 nm is not a nitrogen band. The red MMIA photometer line (OI $3p\,^5P \to 3s\,^5S$) comes, in a cold discharge, from dissociative excitation of O_2: $\sigma = 4.23 \times 10^{-18}\,\mathrm{cm}^2$ at 100 eV (Erdman & Zipf, JGR 92, 1987), i.e. $\sim 3.5 \times 10^{-3}$ excitations per ion pair ($x_{O_2}\sigma_{777}/\sigma_{\mathrm{ion}}$, stated $\times 3$ uncertainty from the degradation-spectrum weighting), heavily quenched ($A = 3.69 \times 10^7\,\mathrm{s}^{-1}$ against collisional quenching at $5.9/9.4 \times 10^{-10}\,\mathrm{cm}^3/\mathrm{s}$ by N_2/O_2 ($O(3p\,^3P)$ TALIF proxies, Niemi et al. 2005); quench factor $\sim 8 \times 10^{-3}$ at 12 km). MMIA sees strong 777.4 nm from hot leader channels (Heumesser et al., GRL 48, 2021, report blue $\sim 3\times$ red for TGF pulses, with the red understood as leader emission), and this simulation contains no leader; its 777.4 nm output is therefore a cold-discharge estimate rather than a leader-emission prediction.

Optical analysis tools. The CLI produces a three-panel figure (lightcurve + ionization driver, source photon rates with the 777.4-nm scenario band, and isotropic-source irradiance at the configured observer, whose range for an overhead observer is derived from the capture's emission-weighted source altitude) plus a companion `.optical.csv` with key intermediates (pair rate, altitude, pressure, quench factors, the single `r337` prediction and separate omitted-swarm envelope). The video renderer imports the same `optical_series` calculation for its optical strip.

Radiative transfer to the observer. Modelled, and on by default. `scripts/rrea_cloud_scattering.py` hands the source series to **CloudScat** (the ALOFT campaign's Monte Carlo radiative-transfer code; Luque et al., *Geosci. Model Dev.* **13**, 5549, 2020), kept as an untracked clone in `external/cloudscat_cpp`. The helper owns the upstream URL, build and physical self-test; no repository document pins a separate revision. The coupling uses CloudScat's own `source. type: "list"`: each reduced-CSV sample becomes a sub-source $\{[0, 0, z(t_i)],\ Q_i = r(t_i)\,\Delta t_i,\ t_i\}$, so the emission altitude follows the beam and no linearity or fixed-position assumption is needed. CloudScat returns photons $\mathrm{m}^{-2}\mathrm{s}^{-1}$ *per emitted source photon*, so the observed flux is that timeline times $\sum_i Q_i$; its next-event estimator already carries the direct term through $e^{-\tau}$, so the result *replaces* $1/(4\pi d^2)$ rather than correcting it. Normalisation is pinned by a closed-form self-test: with no droplets and Rayleigh disabled the time-integrated fluence must equal $\sum_i Q_i/(4\pi d^2)$ within the self-test tolerance.

The size of the correction depends steeply on how deep the source sits below cloud top, which is therefore a flag (`--cloud-top-km`) carried on every figure title rather than a buried constant.

Against a top at or below the emission altitude the source is effectively outside the liquid-water cloud, the direct beam survives, and both the lag and the FWHM change little. A source buried many Mie mean free paths below the top instead loses its direct beam and develops a delayed, broadened tail. The simulation contains no cloud state: any quoted lag, width or peak must name the assumed top and be recomputed from that capture. Note there is no inequality between the scattered and free-space fluences: scattering redirects photons that would otherwise have missed the observer entirely, so the cloudy value legitimately exceeds the bare line-of-sight one. Because the figures share one source-time axis across gamma, optical and radio, much of that tail falls past the right-hand edge; the panel title states what fraction of the scattered fluence the window actually holds. `--no-cloud-scattering` restores the free-space-only panel.

Stated limitations. A single mean emission altitude per time sample (beam mean z), which suppresses the cascade's vertical quench and transport variation; no hot-leader channel, so the 777.4 nm trace is a cold-discharge estimate; the 777.4 nm per-pair yield is a 100 eV cross-section ratio, not a degradation-spectrum integral ($\times 3$ band on the figure). Sub-cutoff radiated energy is never re-deposited as ion pairs (§7); the script reports that energy separately; it does not quantify the optical yield that subsequent transport of those omitted photons would produce.

11 Model Scope and Known Model-Form Uncertainties

The production model is an *axisymmetric (RZ, $m = 0$), zero geomagnetic field* RREA model with one supported field equation: finite-speed Maxwell TM (Section 8.3). The following limitations are model-form or numerical uncertainties.

- **Geometry and fields.** RZ $m = 0$ cannot represent lateral avalanche displacement, filamentation or azimuthally localized feedback. The self-consistent B_θ IS gathered to the charged particles and applied by a Boris rotation (`RreaChargedInterleave.H`); what is omitted is the geomagnetic field, which a horizontal component would need $m \neq 0$ to represent. That omission need not be small: a representative 50 μT geomagnetic field gives a 1-MeV gyroradius of $\sim$95 m, comparable to an avalanche length. This is a model-form limitation of the $m = 0$ geometry, not an assumed-small correction.
- **Temporal discretisation.** The charged transport is first order in the macro step: the microchord drifts on the pre-kick direction, hazards and drag are frozen per segment (the first at the explicit-midpoint energy $K_n + (W - S(K_n)L)/2$, so the runaway threshold energy does not shift), and Goudsmit–Saunderson deflection is applied at the chord end. Quantitative rates therefore need a timestep ladder in the same physical regime; a bound from an earlier source or field history is not automatically transferable. This is a numerical-resolution knob and needs no new table bundle.
- **Low-energy fluid closure.** Production uses the dry-air `en_table_flux_v1` mobility/diffusion tensor and `en_table_attachment_v1` attachment from `electron_flux_mobility_attachment_diffusion.csv`. The table is interpolated in E/N and supplies flux mobility, longitudinal/transverse diffusion, and a source-anchored dry-air attachment rate. Positive and negative ions retain lumped reduced mobilities (`config/rrea_defaults.json`). Electron–ion and ion–ion recombination are separate reaction channels; detachment is

implemented but deliberately zero. The v3 table joins Milloy's zero-field mobility and Hegerberg–Reid drift data to de Urquijo dry-air transport through 60 Td; Morrow–Lowke supplies separate density-linear/two-body and density-squared/three-body attachment above the measured 3–30 Td three-body band, and below 1 Td the three-body rate is the thermal one with the Kossyi electron-temperature dependence (the Morrow–Lowke power law diverges as $E/N \to 0$). Its positive 10^{-6}-Td endpoint is the measured zero-field asymptote. The remaining model-form uncertainty is room-temperature dry-air data and two lumped ion species in cold cloud air; above 60 Td is forbidden because the fluid has no impact-ionisation channel, which reaches 8 % of attachment there. Shimamori–Fessenden (1981, doi:10.1063/1.440853) measured distinct, partly opposing temperature dependences for O_2- and N_2-assisted thermal attachment at $p_{O2} < 10$ and $p_{N2} < 60$ Torr. Those thermal, low-partial-pressure measurements do not supply a validated correction for the heated swarm in this 203–266 K column. The measured room-temperature anchors are retained; MCAIR's O_2-assisted-only channel does not replace them. A matched MCAIR field-collapse diagnostic at $N = 8.8 \times 10^{24}$ m^{-3}, 237 K (two 512-electron ensembles, 28 Td) separates momentum from energy relaxation: after an abrupt switch-off the flux is consistent with zero in the first 0.25-ns sample, whereas the mean energy falls below 0.04 eV after 27–29 ns. After a 20-ns ramp it takes a further 24–26 ns. The conservation residual is below 8×10^{-13} relative. These are direct kinetic histories, not energy/power estimates. They do not validate a universal relaxation time or a reaction closure using only mean energy. The recorded global electron-weighted fields vary on scales down to 1.85 μs, but cannot resolve faster local collapse. With incomplete attachment chemistry, an energy-moment correction remains unsupported; the existing measured closure is retained.

- **Macroparticle resolution.** Source weights and the controller target are run inputs. Exact cell-charge constraints can leave the live count above target when no support group has removable null-space capacity, so the hard ceiling permits those sparse supports without corrupting deposited charge; sampled interactions therefore enter the nonlinear $n_e \to \sigma \to E$ loop in finite-weight packets. Quantitative screening/collapse claims therefore require a population-target convergence ladder.

- **Transport cutoffs.** Bremsstrahlung photons below the table configuration's tracked cut are continuous drag: their backscatter, upstream feedback seeding and displaced deposition are absent (largest exposure: the escaping spectrum below the cut and soft-photon-derived optical inputs). The run records the continuous soft-bremsstrahlung energy so this missing tracked-photon share can be measured directly; it is radiated away rather than re-deposited as ion pairs. Hard-Møller secondaries below the table configuration's threshold are restricted stopping: straggling, displaced knock-ons and their runaway-promotion probability remain model limits outside the 28-Td cutoff study in Section 12. The *transport* cutoff can vary on this bundle down to its 1 keV charged-energy floor, the default. A separate state policy demotes electrons at or below the hard-Møller threshold when $|E| \le S_{\mathrm{ensemble}}(K)$: even a field-aligned electron cannot overcome the mean drag there. A cutoff ladder below 50 keV therefore probes positrons and electrons in stronger fields, not that already-demoted band. An override cannot reach this bundle's 50 keV hard-Møller threshold: the engine aborts unless the charged cutoff is strictly below the table configuration's `hard_moller_secondary_threshold_eV`, which is 50 keV, because hard secondaries born at or below the cutoff would be counted once as continuous drag and once as a dropped particle. Do not confuse this with photon cutoff, which is derived as 50 keV from the table configuration's `brems_absolute_cutoff_eV`: they are different cutoffs on different species. A charged cutoff at or above 50 keV therefore needs a

bundle whose hard-Møller threshold sits above the new value. The two `N_*_ge_100keV_real` columns report the surviving near-cutoff band on every run; use them to size and interpret the ladder in that run's field.

- **Positron annihilation timing.** A positron falling below the transport cutoff is thermalized and annihilated in the same step: there is no thermal positron lifetime and no positronium. The omitted lifetime is density- and positronium-dependent and can be long compared with a timestep or an avalanche e-folding. The approximation can therefore advance annihilation-photon production and must be tested when photon timing or feedback is the observable.

- **Radial taper.** The taper is a declared finite storm-charge model, not a boundary trick; its fringe is fully simulated. The driver audits each resolved field against the local runaway threshold, and recorded energetic tails can enter the taper, so the fringe cannot be dismissed as unsampled.

- **Observables.** Optical irradiances are scenarios conditioned on the assumed cloud top and on the emission model; 337 nm predicts measured deposition fluorescence with a separate swarm assessment, and 777.4 nm remains a leaderless cold-RREA estimate. The standard waveform panels use reduced Huygens observer fields when available. The fallback z-resolved line-source model uses dM/dz but collapses radial currents to the axis; it is explicitly labelled. Its 1% moment check bounds a source approximation, not charge conservation: kinetic bins use survivor velocities while Maxwell uses step-averaged net chords, including terminal particles. Samples outside the supported retardation window are blanked. Independently of geometry, the EFCM prediction is conditional on the flat plate's along-heading orientation (Kochkin et al. 2021): at exact nadir the instrument reads zero whatever the source does, and the nadir prediction lives in the step read by the LIP field mill instead. At a 100 km ground receiver the axial compact-source error is instead small, $(L/R)^2 \approx 0.01$ for the production domain height, but the radial extent sets a corner $c/(2\pi\,\mathrm{rms}_r \sin\theta)$ (26 kHz for the E120 capture) above which the collapsed dipole misses the broadside phase spread of the $\sim$2 km-wide current. The figure therefore overlays the dipole only on bands that start below this corner; above it the surface integral is the result.

Not modelled: Xu's discriminating band ratio. Xu et al.'s headline diagnostic is the $2\mathrm{PN}_2/1\mathrm{NN}_2^+$ intensity ratio, which separates production mechanisms (their Table 2: 0.74, 1.10, 5.06 for RREAs at 4.3, 12.5, 18.8 kV/cm, against 1.58 for thermal runaway at a leader tip and 101 for the streamer zone). Computing it needs the 391.4 nm $1\mathrm{NN}_2^+$ band, which this pipeline does not produce. That is a deliberate scope choice, not an oversight: the target instrument is ASIM/MMIA, whose photometers are 337 nm, 777.4 nm and 180–230 nm, and 391.4 nm is not among them. A comparison against Xu's mechanism ratio, or against a ground/TARANIS 391.4 nm photometer, would require adding that band first. The 180–230 nm LBH channel is likewise not modelled.

12 Checking the Physics: Avalanche Multiplication vs. Coleman–Dwyer

Focused tests establish specific implementation properties; they cannot prove that the model matches nature. For RREA physics the community yardstick is the *avalanche length* λ: the distance over

which the runaway electron population multiplies by a factor e. Coleman and Dwyer distilled large independent Monte Carlo campaigns into one empirical formula,

$$\lambda_{\mathrm{CD}} \;=\; \frac{7300\ \mathrm{kV}}{E_0 - E_{\mathrm{th}}\,\rho},$$

where E_0 is the applied field, E_{th} the effective runaway threshold, and ρ the air density *relative to the density at which E_{th} is quoted*. That reference is 1.293 kg/m^3 (dry air, 0 °C, 1 atm), not the table reference of Section 7.1 and not a sounding's own surface density. It is fixed independently of any simulation: Dwyer's break-even field, 218 kV/m, is where the field force equals the minimum total collision drag, and NIST ESTAR puts that minimum at ≈ 1.66 MeV cm^2/g for dry air, so the two are consistent only near 1.31 kg/m^3. Anchoring E_{th} instead at a sounding's own surface density raises the threshold, and that error both shrinks the above-threshold interval and narrows the margin inside it, so it suppresses the predicted multiplication. Two threshold values bracket the reference: the 276 kV/m intercept of the C&D avalanche-length fit and Dwyer's 284 kV/m onset field. This project benchmarks against *both* and reports a descriptive ±10 % consistency band on the avalanche length (the quantity C&D fitted) for each; with $M = e^S$ that is a field-dependent band on the multiplication. These comparison labels are not software pass/fail labels.

The benchmark must compare like with like. Reduced constant-field and mono-particle probes test components but are not C&D results. Every campaign comparison runs through `scripts/run_rrea_profiled_video_capture.py`, with the real altitude profiles and compact video diagnostics still enabled. The dedicated source is a cohort of simultaneous, unit-weight electrons on a fixed source disk (cohort size `CD_INITIAL_SEED_COUNT` in `scripts/rrea_cd_physics.py`, disk radius `channel_radius_m` in `config/rrea_defaults.json`). Their energies are deterministic stratified quantiles of the 7.2 MeV exponential, conditioned to the certified table range; independent realizations use distinct transport RNG seeds.

Population peaks and time-series peak/trough ratios are not multiplication observables. The engine instead accumulates cumulative upward and downward electron weight at physical planes, and the offline report forms the signed time-integrated flux

$$\Phi_{\mathrm{net}} = \Phi_{\uparrow} - \Phi_{\downarrow}, \qquad M = \Phi_{\mathrm{net}}(z_{\mathrm{end}})/\Phi_{\mathrm{net}}(z_{\mathrm{source}}),$$

so equal-state recrossings cancel. The primary selection is $K \geq 1$ MeV: above 1 MeV the avalanche spectrum does not depend on the field, so the counted fraction is the same on every plane and the flux ratio is the avalanche growth. The *locally runaway capable* selection (gathered field above the ensemble mean drag, including the mean hard-Møller and tracked-bremsstrahlung losses) is reported alongside; its energy window moves with the field along the profile and biases M by several percent where the field approaches threshold. The all-tracked channel is retained as the exact topological cancellation audit.

The common physical interval starts at the injection altitude and ends at the first upper crossing of $|E| = 276n$ kV/m. Both denominator choices use those same endpoints:

$$S_T(z) = \int_{z_{\mathrm{in}}}^{z} \frac{E(z') - Tn(z')}{7300\ \mathrm{kV}}\,\mathrm{d}z', \qquad T \in \{276, 284\}.$$

Profile knots and crossings are integrated exactly. Endpoint compatibility and a post-equilibration spatial fit are reported separately; fields that do not meet the configured equilibration and plane-coverage criteria remain endpoint-only.

The exact interval above is the analytic context of the quoted profile table, but the diagnostic planes sit half a cell inside its two ends, so a measured flux ratio is compared with the same integral taken between those plane positions. A measured flux is never extrapolated to an unobserved endpoint, and a supplied plane list never chooses its own interval.

The implemented campaign is an early-time all-physics comparison: photon and positron transport, charged secondaries, the fluid and the self-consistent field remain enabled. The short observation window can reduce the contribution from long feedback loops, but it is not a feedback switch; high-field excess multiplication may be genuine feedback and is reported as a deviation rather than used to recalibrate transport; late-window flux growth is reported but never drops a realization. In a unit-weight realization, a strict-ceiling abort means the count exceeded five times the 276 reference. A complete realization is the statistical unit, and cohorts with different seed statistics or population targets are not pooled.

Exact fields, durations, cohorts and eligibility rules live in `scripts/rrea_cd_physics.py`; the planner consumes those values and `scripts/report_rrea_coleman_dwyer_production.py` performs the reduction. This section defines the physical observable, not a reference result.

Hard-secondary cutoff study at 28 Td (27 October 2026). The production 50-keV ensemble drag balance is 28.298 Td using the actual Geant4 mass-density and engine number-density conventions. The narrow gap to the sampled E125 maximum is not itself a convergence argument. Restricted ionization stopping, hard rates and secondary CDFs were therefore regenerated together at 50, 25 and 12.5 keV with Geant4 10.7.4. The partition $S_{\mathrm{restricted}} + \Sigma\langle W\rangle = S_{\mathrm{unrestricted}}$ closes within 0.05% for electrons and 0.5% for positrons. These are numerical reference-model checks; the cut-dependent Geant4 interpolation also affects the low-energy positron CDF, and is not a measured spectrum.

Six independent, unresampled 1000-electron cohorts per cut traverse a 28-Td layer with a 9420–9800 m-MSL plateau and 20-m edges, retaining all transport and field processes. The common 9607.336–9757.336 m interior interval starts beyond 2.5 reference e-foldings. At 1.75 μs, after the primary passage and before appreciable return feedback, the measured signed $K \geq 1$ MeV flux ratios are shown with whole-domain ionization and the recorded current moment $\mathcal{J}_z = \int J_z\, dV$:

Cut (keV)	Interior multiplication	Pairs at 2 μs (10^{11})	$\lvert\int \mathcal{J}_z\, dt\rvert$ at 2 μs (10^{-10} C m)
50	23.118 ± 0.118	5.160 ± 0.101	6.152 ± 0.120
25	23.093 ± 0.115	5.128 ± 0.190	6.127 ± 0.222
12.5	23.033 ± 0.120	5.031 ± 0.147	6.023 ± 0.174

Intervals are 95% Student intervals over realizations, not particle-count Poisson errors. The C&D reference multiplications on this same interior interval are 27.605 (276 kV/m) and 26.110 (284 kV/m): the coarse-grid 50-keV run's logarithmic growth is respectively 0.9465 and 0.9627 of those references. Lowering the cut does not move this systematic difference toward either reference. The table uses 128×512 cells and 2.5-ns steps; its whole-domain pair/current errors are statistical only.

Mesh checks at 12.5 keV used 256×1024 cells with 1.25-ns steps, then 256×2048 cells (4.88 m axial spacing) at the same timestep. From 9.77 to 4.88 m, the interior flux ratio changes by -1.23% (0.40% in logarithmic growth), pairs by -0.17%, and integrated current by $+2.05\%$. The earlier

check changed both spatial resolutions and the timestep together, changing current by 6.35%; their effects are not isolated. At 4.88 m, lowering 50 to 12.5 keV changes interior multiplication by +0.33%, pairs by −1.49%, and current by −1.06%. These refinement comparisons are single realizations; their stochastic differences are not pure discretization errors. All compared observables use the same physical planes and times, rather than the mesh-dependent end planes.

The retained hard-secondary threshold is therefore 50 keV for the observed near-28-Td regime. No resolved cutoff effect warrants a lower threshold; bulk pair/current precision in this study is a few percent, not 1%. This does not validate the handoff throughout the fluid table's 60-Td domain. The runtime charged coefficients, secondary distributions, measured W and energetic scattering remain unchanged. The extraction tool supplies reference CSVs; assembling runnable bundles remains a separate manual step.

13 Fail-Closed Operation and Focused Validation

Invalid inputs and failed numerical invariants stop the affected operation:

- source preparation aborts if WarpX's loop shape changed;
- startup aborts on any species, table semantics, model-string, or cutoff mismatch;
- `RreaInteractionTables::Load`, the one owner of table semantics, validates the structures and values the engine actually consumes;
- transport aborts on optical-depth guard violations and ledger (charge/energy conservation) violations;
- when enabled (production runs and the default bounded-population C&D check), `adaptive_resample_v1` thins the population toward `rrea.population_target_electron_macros` within a hysteresis band. Within each common order-1 CIC support, stochastic null-space weight moves preserve all four deposited cell charges exactly in every realization while keeping $E[w]$ for every particle. The controller plans globally over valid survivors plus staged charged candidates and preflights both plans, the stored-precision CIC moments, the physical ledgers and the three ceilings before either mutates a live weight, on deterministic global support groups so capacity and the geometric floor are decomposition-invariant. Exact CIC preservation means the nonlinear charge–conductivity–field loop receives no resampling charge impulse or correction current; particle and spectral observables stay stochastic, unbiased in their weight-linear estimates and bounded in variance only by a macroparticle ladder (§11);
- the checkpoint reader refuses an inconsistent payload and a detected rank-count or reduced-schema change without an explicit restart epoch;
- focused native smokes and `scripts/test_*.py` print an obvious PASS/FAIL; the module index (§15) names the pin of every physical process. Independent anchors include the ESTAR stopping-power comparison in `native_amrex_rrea/src/RreaScatteringSamplerSmoke.cpp` and the primary-source fluid-closure checks in `scripts/verify_fluid_closure_anchors.py`. C&D results run the real profiled production path; reduced probes remain component tests.

14 Execution backend and measured GPU scaling

The experimental CUDA path shares the CPU physics primitives. Kinetic histories, field/fluid updates and population kernels have device paths; ordered source replay, insertion and MPI orchestration still involve the host. Some Maxwell arrays and work remain replicated. On Olivia, one MPI process owns each GH200, including within a single node. The Olivia wrapper defaults to GPU kinetics, fluid and Maxwell, but the production deck and the retained GH200 layout run kinetics on the host (`rrea.kinetic_backend=cpu`, 48 transport threads) with GPU fluid and Maxwell. See `GPU_PORT.md` for implementation limits and `OLIVIA_GPU.md` for build, launch, configuration and transfer commands.

Measured workload (19 September 2026). Historical baseline: branch `gpu-cuda`, numerical source `c77e83b`, double precision, CUDA 12.6 and OpenMPI 5.0.3. Replay from `chk042480` used the same seed schedule, 2000×1000 mesh, $\Delta t = 2.5$ ns and peak ambient field 120 kV/m. Every GPU count used the same MPI+CUDA executable, one host thread per rank, managed memory, sorting/tiling disabled and ordinary reduced output; no video frame was due. Two 20-step replays per count excluded the first five steps, timing steps 42486–42500. Each simulation occupied one node; different counts used different nodes. CUDA device identities confirmed one distinct physical GPU per rank.

Table 1: Historical simulation timings and current radio performance. Lower time is better.

Machine	Resources	Time (s/unit)	Relative speed
Round 7 at `7af592c`, one/two GH200s (not remeasured on the current engine)			
Olivia	1 GH200 + 48 host threads	1.671	4.26×
Olivia	2 GH200s + 96 host threads	**1.556**	4.57×
Historical baseline (not remeasured after optimization)			
Local	6 CPU cores	7.112	1.00×
Olivia	16 CPU cores	3.491	2.04×
Olivia	32 CPU cores	1.866	3.81×
Olivia	64 CPU cores	2.184	3.26×
Olivia	1 GH200	9.622	0.74×
Olivia	2 GH200s	6.948	1.02×
Olivia	3 GH200s	4.917	1.45×
Radio, 28 October 2026: retained Numba, R9, 12 threads, s/sample			
Olivia	Ground observer	0.017017	—
Olivia	Aircraft observer	0.016882	—
Olivia	Nadir observer	0.000899	—

R7 rows: six replays each, CPU kinetics with GPU fluid/Maxwell. Historical rows: two replays each, one MPI rank per CPU core or GPU. All use the same mature checkpoint and scored steps. Ratios use the historical local CPU time; mixed code versions are not a current-code scaling comparison. Radio rows use R9's fully active windows, described below: staging and kernel time per observer sample, excluding initial input, geometry and JIT compilation. They are not full-run elapsed-time predictions. Simulation rows use seconds per timestep; radio observers have different workloads, so no cross-observer speedup is assigned.

Historical rows: the local CPU is a Ryzen 9 7900 under WSL2; Olivia CPU rows used one AMD EPYC 9745 node, one MPI rank per allocated core and GCC 14.2; GPU rows used GCC 13.3. They are practical backend comparisons, including host architecture and compiler differences, not isolated accelerator ratios. 64 cores took 17% more time per step than 32 (node placement was not held fixed), and two and three GPUs reached 69% and 65% efficiency. Short windows and different nodes limit precision; these are not whole-discharge projections or evidence of linear scaling.

At step 42500, the six GPU replays had 497535–497742 native particles, field fraction 0.859948, relative Gauss residual about 3.00×10^{-8} and charge-continuity residual 1.20–1.68×10^{-16}. These support comparable workload and short-run numerical consistency, not full physical validity or convergence. The production builds passed 28 local native checks and 25 CUDA checks. CPU-to-GPU restart was exercised by these replays; a separate GPU checkpoint continued on the local CPU. Job IDs and raw evidence locations are in `../JOURNAL.md`.

Optimization and scaling rounds. Each R1–R7 confirmation used six adjacent, counterbalanced pairs on one allocation, 20 steps per run with five discarded. Confidence intervals use the paired run log-ratios, not correlated individual steps. Round 2 holds Round 1's hybrid setting fixed: twelve host transport threads, GPU fluid and Maxwell.

Each retained simulation round passed 76 comparisons against eight matched local CPU continuation seeds (1% bulk and 5% sampled energy-band/feedback bounds, paired 95% intervals inside them), with charge continuity at most 2.4×10^{-16} and Gauss residual 3.0×10^{-8}. This checks numerical and statistical equivalence on the sampled workload, not rare tails or model validity. R2 and R6 cache the last one and two valid accumulator boxes; recipient choice, grown-box precedence, accumulation order and RNG streams are unchanged. R4 changes only the transport-thread count (72 threads added 1.1%). R3 (transfer-buffer reuse) and R5 (`GOMP_SPINCOUNT=0`) missed the 2% gain threshold and were removed. R7 keeps one GH200 as the default: two GPUs gave 7.4% more throughput for 86% more GPU time per step. Profiles, replays and job IDs per round are in `../JOURNAL.md`.

Table 2: Optimization and scaling rounds. Times are seconds per step, except R8–R9 (per observer sample).

Experiment	Before	After	Speedup	Paired 95% CI
R1: host kinetics, 12 threads	10.198	9.172	1.112×	[1.110, 1.114]
R2: last-valid-box cache	8.439	2.051	4.115×	[4.107, 4.123]
R3: metadata reuse	3.007	3.041	0.989×	[0.967, 1.011]
R4: host threads $12 \to 48$	2.049	1.802	1.137×	[1.111, 1.164]
R5: zero worker spin	1.830	1.828	1.001×	[0.988, 1.014]
R6: two-entry valid-box cache	1.868	1.705	1.095×	[1.063, 1.128]
R7: GH200s $1 \to 2$	1.671	1.556	1.074×	[1.066, 1.083]
R8: compiled radio sum	0.30794	0.06159	5.000×	—
R9: C++ ground	0.017017	0.016460	1.034×	[1.008, 1.060]
R9: C++ aircraft	0.016882	0.016406	1.027×	[0.979, 1.078]
R9: C++ nadir	0.000899	0.000905	0.993×	[0.973, 1.014]

R1, R2, R4 and R6 were kept; R3 and R5 were rejected. R6 saves 8.7% of step time. R7 measures scaling; one GPU remains the default. R8: NumPy/1 thread to Numba/12 threads, twice on 80 samples from the first 80000 E125 records (stride 1000), including geometry and staging but excluding JIT and disk input; no confidence interval. Production sampling was unchanged. E/B differences were below 6×10^{-16} of peak; 48 checks passed. Indexing took 84.65 s for 1179878 records; strided reads had not finished after 30 minutes. R9: Numba 0.63.1 before, GCC 14.2 C++/OpenMP after, with fixed node order and no fast-math or FMA contraction, on the same EPYC 9745 core sets. The screen used 1/4/12 threads, all three observers, startup/middle/end windows, stride 4 and 256 azimuth nodes on the 1992×1424 surface. Rows show 12-thread means over the fully active middle/end windows (256 samples each), including staging, allocation and calls. Six alternating pairs per window give the paired-log intervals after summing the two window times within each repeat. Both backends passed 56 checks; measured value/derivative/E/B differences were zero. Peak process RSS was 11.6 GiB. C++ saved only 3.3%/2.8% for ground/aircraft and was rejected against the required 50% saving; no full reruns or production C++ were added. Control timings differ across allocations: compare within each row, and do not multiply speedups or interpret these intervals as across-node uncertainty.

15 Physics-Module Index

Repository requirement (`AGENTS.md`): every physical process lives in ONE named module with its governing equation or model citation at the top, reviewable in isolation and pinned by a native smoke or committed test. This index is the map for reviewing them one by one: process: module; pin: test. Native smokes are ctest targets under `native_amrex_rrea`; Python pins live in `scripts/test_*.py`. Two pins below are neither: `rrea_warpx_actual_continuity` and `rrea_warpx_transport_apply` are injected into the prepared WarpX tree by `scripts/prepare_rrea_warpx_source.py` and need a built `app_rz`, so they are build-scale, not minutes-scale.

Kinetic transport

- Chord geometry (cylinder and axial-face exits, in-domain span): `RreaTransportPath.H`; pin: `rrea_transport_path_smoke`
- Gauss–Legendre quadrature (the 2-node rule, exact for a cubic): `RreaGaussLegendre2.H`; pin: `rrea_gauss_legendre2_smoke`
- Geometric density path ($\tau = \sigma_{\mathrm{stp}}(E) \int \chi \, \mathrm{d}\ell$ with $\chi = \rho/\rho_0$; midpoint χ over the in-domain span, relative error $\leq \frac{1}{24}(h/H)^2$; depth-to-position inversion exact): `density_path_m` / `fraction_at_density_path_m` / `max_uniform_subcycle_density_path_m` (`RreaMaterialPath.H`), with $\chi(z)$ from `TransportDensityView` (`RreaKineticEnvironment.H`, shared with the host); pin: `rrea_transport_path_smoke` + `rrea_charged_interleave_smoke`
- Line-integral field work along a microchord ($W = q \int_a^b \mathbf{E}\cdot\hat{d} \, \mathrm{d}\ell$ and mean $\mathbf{E}, \mathbf{B}$ from the same 2-node Gauss–Legendre rule; exact within one interpolation cell for a meridional chord, approximate across cell faces or along azimuthal chords): `path_field_work_and_average` (`RreaPathFieldWork.H`); pin: `rrea_gauss_legendre2_smoke` (cubic exactness) + `rrea_charged_interleave_smoke` (the half-kick consumer)
- Field half-kick + rotation: `RreaApplyChargedFieldHalfKick` (`RreaChargedInterleave.H`) + `RreaLorentzPush.H`; pin: `rrea_charged_interleave_smoke` (work-only energy, zero floor, charge-sign reversal)
- Hazard optical depth + inversion: `RreaResolveRemainingSegmentHazard` (defined in `RreaParticleInteraction.H`, driven from `RreaChargedInterleave.H`); pin: `rrea_charged_interleave_smoke`
- Event-interleave sequencing (midpoint first freeze, refreeze, null events): `RreaRunChargedEventInterleave` + `RreaChargedMidpointFreezeEnergy`; pin: `rrea_charged_interleave_smoke`
- Substep skeleton: `RreaRunChargedSubstepInterleave`; pin: same
- Continuous drag + ion-pair partition (the cutoff point on the chord by exact inversion of the same density path; ion pairs at the segment midpoint): `RreaPlanContinuousDragSegment` → `RreaContinuousDragSegmentPlan` + `RreaPartitionContinuousIonPairs` (`RreaChargedInterleave.H`); pin: interleave smoke

- Local-energy ion-pair deposit ($N = wE/W_{\text{air}}$, $W_{\text{air}} = 33.97$ eV, ICRU 31): `RreaIonPairWeightFromLocalEnergy` (`RreaParticleInteraction.H`); pin: `rrea_secondary_group_smoke`
- GS elastic application (one unit for all charged paths, RNG-key policy): `apply_gs_elastic_scatter` (`RreaKineticStep.H`); pin: the ANGLE SAMPLER it calls is pinned by `rrea_scattering_sampler_smoke`; its application by `rrea_warpx_transport_apply`
- Charged discrete channels: phase-space gates, frozen competing rates (the STP inverse-length table where the gate is open, exactly zero elsewhere), and complete hard-Møller / hard-Bhabha / bremsstrahlung final states (sampled kinematics, parent post-state, secondary, material sources, group closure; the tracked-brems channel opens only above the photon transport cutoff, for both species); `RreaChargedChannelStpRate`, `RreaChargedCompetingStpRates`, the two-body knock-on lab angle $\cos^2\theta = T(T_0 + 2mc^2)/(T_0(T + 2mc^2))$ for both outgoing leptons (`RreaMollerCosThetaLab`, `RreaChargedInterleave.H`), `RreaApplyHardMollerFinalState` / `RreaApplyBhabhaFinalState` / `RreaApplyBremsFinalState` (`RreaChargedFinalStates.H`); pin: `rrea_charged_interleave_smoke` (kinematics, recoil) + `rrea_scattering_sampler_smoke` (tracked/local brems ledger) + `rrea_secondary_group_smoke` (group conservation) + `rrea_warpx_transport_apply` (assembled step)
- Photon discrete channels: Compton, photoelectric absorption, nuclear and triplet pair production, each booking its energy flow through the one owner of the expected split, with one tracked / sub-cutoff cascade emission for Compton recoils and photoelectrons (`RreaEmitPhotonFreedElectron`); `RreaApplyComptonFinalState` / `RreaApplyPhotoelectricFinalState` / `RreaApplyPairFinalState` (`RreaPhotonFinalStates.H`) consuming `RreaComptonPhotonEnergyFlow` / `RreaPhotoelectricPhotonEnergyFlow` / `RreaPairPhotonEnergyFlow` (`RreaParticleInteraction.H`); pin: `rrea_scattering_sampler_smoke` (samplers) + `rrea_secondary_group_smoke` (group conservation) + `rrea_warpx_transport_apply` (assembled step)
- Annihilation booking (in flight / at rest; tabulated in-flight rate, analytic Heitler final state `sample_annihilation_photon_energy_fraction`, `RreaInteractionPhysics.H`): `spawn_positron_annihilation` (`RreaKineticStep.H`); pin: `rrea_scattering_sampler_smoke` (Heitler sampler) + `rrea_warpx_transport_apply`
- Demotion gate + state policy: `RreaEvaluateElectronStatePolicy` (`RreaParticleInteraction.H`); pin: secondary-group closure smoke
- Charged and photon histories (one each for live and newborn particles; a secondary born at parent-chord fraction f flies the remaining $(1 - f)\Delta t$ on its root ancestor's RNG stream within its own lineage): `AdvanceChargedHistory` / `AdvancePhotonHistory` with adapters `Charged` / `Photon` / `AdvanceNewborn` (`RreaKineticStep.H`), clock `RreaNewbornElapsedStepFraction` (`RreaParticleInteraction.H`); pin: `rrea_kinetic_engine_smoke` (exit outranks demotion, live and newborn substep counts agree) + `rrea_charged_interleave_smoke` (clock) + `rrea_secondary_group_smoke`
- Terminal/escape ordering: `RreaResolveChargedTransportTerminal`; pin: interleave smoke

- Realised and ensemble drag (collision + soft radiative; plus the mean hard-Møller and tracked-brems losses for force balance): `ChargedRealizedDrag` / `ElectronEnsembleDrag` (`RreaInteractionPhysics.H`), also served by the `drag` mode of `rrea_interaction_table_probe`; pin: `rrea_scattering_sampler_smoke` (ESTAR minimum) + `rrea_warpx_transport_apply`
- Secondary-group charge/energy ledger: `RreaParticleInteraction.H` group validation; pin: `rrea_secondary_group_smoke`
- Transport transaction log (ordered replay for material sources, charged deposits, the Compton/photoelectric/pair records and positron annihilation; associative scalar tallies in a per-chunk `RreaEventLedger` folded once by `RreaFoldEventLedger`): `RreaTransportLog.H` + `RreaEventLedger.H`; pin: `rrea_warpx_transport_apply` + `rrea_warpx_actual_continuity`
- Counter-based RNG streams (one `splitmix64` mixer, `RreaKineticSupport.H`, behind the counter RNG, the rejection-sampler stream and the CIC support keys): `RreaRng.H` / `RreaUniformStream.H`; pin: `rrea_rng_endpoint_smoke`
- Species transport phases (one tile loop for electrons, positrons and photons: GPU batch or host chunks, side effects merged in chunk order): `TransportStep::TransportSpecies` (`RreaSpeciesTransport.cpp`); pin: `rrea_warpx_transport_apply` + `rrea_warpx_actual_continuity`
- Secondary transaction commit (collective charge/energy closure of every staged group, then the newborn advance and the terminal writeback): `TransportStep::DrainAndCommitSecondaries` (`RreaSecondaryCommit.cpp`); pin: `rrea_secondary_group_smoke` + `rrea_warpx_actual_continuity`
- Per-step Monte Carlo entry point (builds the `TransportStep` context, runs the four phases in order): `RreaParticleInteraction::Apply` (`RreaParticleInteraction.cpp`); pin: `rrea_warpx_transport_apply`

Interaction tables and samplers

- Load + semantic validation + enum-indexed access (one `RreaTable` id per table, bound once by `BuildKineticStorage()`): `RreaInteractionTables::Load` (the one owner of table semantics); pin: `rrea_interaction_tables_smoke`
- RREATBL container decode (header gate, named columns, un-shuffle, finiteness): `RreaTableContainer.H`; pin: `rrea_table_container_smoke` + a Python round-trip pin
- Log-log rate interpolation (no clamp/extrapolate): `RreaInteractionTable::Interpolate`; pin: endpoint/midpoint pins plus Geant4 holdouts (1% relative error, 10^{-7} m^{-1} absolute rate error, and 2×10^{-12} relative energy uncertainty at model jumps)
- Charged-ionization reference extraction: `secondary_quantiles` in `tools/generate_charged_tables/charged_tables.cc`, $F(W) = 1 - \Sigma(W)/\Sigma(c)$ (eV, m^{-1}); pin: analytic W^{-2} quantiles via `--self-test`.

- Photon channel partition: `SamplePhotonChannel`; pin: pinned vs `PhotonProcessInverseLengths`
- GS angle sampler (endpoints are BOUNDS): `sample_gs_angle_cos`; pin: `rrea_scattering_sampler_smoke`
- Screened-Rutherford single elastic: analytic-quantile sampler; pin: pinned to closed form
- Wentzel-VI branch above the 100 MeV transition: `SampleGsCosTheta`; pin: the sampler smoke's single-scatter chain sweep at 0.15–5 GeV
- Brems quantiles + bounded Modified–Tsai angle: conditional CDF + `SampleBremsstrahlungCosTheta`; pin: energy holdouts, truncated-density moments, endpoint support and e-/e+ event closure in `rrea_scattering_sampler_smoke`
- Møller/Bhabha secondary CDF: per-species samplers (endpoints $T_{cut} \dots E/2$ for identical electrons, $T_{cut} \dots E$ for Bhabha); pin: sampler smoke + holdout
- Pair fraction + final-state bank replay: loader/validator + `SamplePairFinalState` (replay, validates the loader); pin: norm, coverage, $\langle\gamma\theta\rangle$, charge-symmetry pins
- Photoelectric subshell + Sauter–Gavrila angle: subshell sampler + angle unit; pin: closure, K-edge, angle pins
- Compton (Butcher–Messel / Klein–Nishina): `RreaInteractionTablesView::SampleComptonKleinNishinaFinalState`; pin: analytic endpoint and KN-moment pins
- Annihilation spectrum (Heitler) + CM→lab; pin: spectrum and final-state unit pins
- Stopping/drag accessors + density-ratio scaling (every rate exactly linear in ρ/ρ_0; the STP bundle is frozen once per refreeze and multiplied by the geometric density path): thin table reads, linearity stated at Load; pin: independent Geant4 holdouts at the table density (the scaling is one multiply)

Field solve

- Faraday update (RZ TM): `MaxwellAdvanceB` (`RreaMaxwellTMRZ.H`); pin: `rrea_maxwell_tm_rz_smoke`
- Ampère update + material current: `MaxwellAdvanceE` (face currents only); pin: same (Gauss follows the continuity current; Joule/Poynting closure)
- Silver–Müller boundary: `RreaMaxwellTMRZ.H`; pin: measured residual in the smoke
- Axis regularity + CFL gate: solver loops + `RreaAmrexAdvance.cpp` gate; pin: `rrea_maxwell_tm_rz_smoke` (axis) + `rrea_amrex_advance_smoke` (the two-dimensional Yee CFL refusal)
- Discrete RZ metric (cell/dual volumes, r-weighted divergence): `MaxwellTMRZLayout` members and pointer-only views (`RreaMaxwellTMRZ.H`); pin: closed-domain exact energy conservation + Gauss pins in the same smoke

- Ambient construction (curl-free taper): `RreaFieldInitializer.H` (ϕ/taper) + `InitializeBackgroundState` (`RreaAmrexAdvance.cpp`; face realization, curl $\equiv 0$ verified fail-closed at every start, then committed as the $t = 0$ field); pin: advance smoke + the Python $|f'|_{max}$ identity pin
- Total-current assembly (fluid + kinetic face current in one concatenated allreduce per substep): `AssembleTotalFaceCurrent` (`RreaAmrexAdvance.cpp`); pin: continuity gate exercises it
- Material-continuity cell identity and cancellation scale, $\rho^{n+1} - \rho^n + \Delta t \nabla \cdot J$ in SI: `EvaluateMaterialContinuity` (`RreaMaterialContinuity.H`), shared by the reduction and its witness; pin: `rrea_gpu_primitives_smoke` (analytic RZ divergence, conserved state, unmatched charge, nonfinite state) + advance smoke
- Continuity reference priming (the seed batch's own charge enters ρ^n before any particle moves, so an injection is not read as a continuity violation): `PrimeMaxwellContinuityReference` (`RreaAmrexAdvance.cpp`); pin: `rrea_amrex_advance_smoke`
- Adaptive symmetric material substeps + continuity gate (+ charge-resolution floor): `AdvanceOneStep` selects from source-projected conduction, diffusion and reaction stiffness, applies half-reaction/transport–Maxwell/half-reaction, and time-averages the shared currents for `VerifyMaterialContinuity`; pin: fluid/advance smokes (including neutral source bursts, +mpi4)
- Energy ledger (dual-volume energy, bilinear-mean Joule, exact discrete boundary term, ambient cross-term): `MaxwellFieldEnergy` / `MaxwellJoulePower` / `MaxwellBoundaryPoyntingPower` (`RreaMaxwellTMRZ.H`), assembled in `AdvanceOneStep`; pin: `rrea_maxwell_tm_rz_smoke` (open-step identity to round-off, closed-domain conservation of the mixed-time form `MaxwellMixedMagneticEnergy`)
- Gauss monitor (never a gate): `RecordGaussMonitor` on `DivergenceRZ`; pin: Maxwell smoke Gauss pin
- Huygens surface record + scattered-state persistence (with the cumulative signed continuity residual, optional so older checkpoints still load): `RreaMaxwellTMRZRecord.H`; pin: native record/restart smoke and reader-side layout pins in `test_delivery_physics.py`; the checkpoint reader owns the mesh check
- Cell-centred commit and total-field gather (the gather forms its cell corners from the replicated Maxwell state by the same two-point face average the commit uses): `CommitMaxwellTotalField` (`RreaAmrexAdvance.cpp`) / `RreaKineticFieldView::Gather` (`RreaKineticFieldView.H`); pin: `rrea_amrex_advance_smoke` (bilinear identity, odd-radial axis scale)
- Particle gather stencil: `RreaFieldGatherStencilFor` (`RreaFieldGatherScalar.H`); pin: `rrea_field_gather_stencil_smoke`
- Cell-centred-to-native field handoff (nodal directions average the two bracketing cells; only physical boundaries clamp): `InterpolateCellCenteredToNative` (`RreaFieldHandoff.H`); pin: `rrea_warpx_actual_continuity`

- Air density vs altitude (the ratio every rate, drag, attachment and mobility is rescaled by, §7.1): `RreaAltitudeProfile.H`/`RreaAltitudeProfile.cpp`; pin: `rrea_altitude_profile_smoke`

Low-energy fluid

- Chemistry channels (exact-exponential attachment, pair-loss ODE, detachment): `LowEnergyFluidState` update, per-channel ODE comments; pin: `rrea_low_energy_fluid_smoke`
- Closure coefficient interpolation (piecewise log-linear in E/N, linear coefficient across a zero-onset bracket, explicit range status): `ElectronClosureViewT::Evaluate` (`RreaElectronClosure.H`), used by the host owner and device kernels; pin: `rrea_gpu_primitives_smoke` (analytic coefficients, exact nodes, onset and range policy) + `rrea_low_energy_fluid_smoke`
- en_table closure evaluation (E/N, hold-below, abort-above), evaluated once per substep at the end of the fluid/Maxwell substep: `RreaAmrexAdvance::EvaluateClosureFields` driving `BuildElectronClosureCellFields` + the closure formulas in `LowEnergyFluidState.H`; pin: face-transport + `rrea_low_energy_fluid_smoke`
- Drift face flux (donor upwind) + diffusion tensor: `BuildExplicitFaceTransport`; pin: `rrea_low_energy_face_transport_smoke` (+mpi4)
- FV update + positivity/outflow limiter: `ApplyExplicitCarrierTransport`; pin: same (limited-fraction diagnostics)
- Conductivity assembly: `ComputeConductivity`; pin: fluid smoke
- Mobility law (`absolute` vs `reduced_stp_density_scaled`, and the conversion between the table density ratio ρ/ρ_0 and the reduced-mobility ratio ρ/ρ_L): `MobilityLaw` / `resolve_mobility_law` (`LowEnergyFluidState.H`) and `RreaWarpXCoupling::MobilityDensityRatio` (`RreaWarpXCoupling.cpp`), with $\rho_L = n_L M_{\mathrm{air}}/N_A$ derived in `RreaConstants.H`; model strings and mobility values are validated once, by `LowEnergyFluidState::ValidateCarrierTransportConfig`; pin: `rrea_low_energy_fluid_smoke` + `rrea_low_energy_face_transport_smoke`
- en_table closure *bundle* anchoring (zero-field–60 Td mobility, diffusion and two-/three-body attachment): `scripts/generate_rrea_fluid_closure_bundle.py` + `scripts/verify_fluid_closure_anchors.py`; pin: primary-source anchor check over the shipped CSV
- Vacuum/subnormal rule: `LowEnergyFluidState.H` (bounded-freeze statement); pin: `run_transport_vacuum_and_seam_test` in `rrea_low_energy_face_transport_smoke`
- Reaction charge audit: in-update audit (throws at 512ε); pin: fluid smoke

Deposition and population

- Order-1 cell-centred CIC shape ($x_{\mathrm{cell}} = (x - x_{\mathrm{lo}})/\Delta x + i_{\mathrm{lo}} - \frac{1}{2}$, support $\lfloor x_{\mathrm{cell}} \rfloor$ and the cell above): `cic_axis` (`RreaConstants.H`), shared by the segment deposit, the event accumulator and the population supports; pin: `rrea_segment_deposition_smoke` + `rrea_event_accumulator_smoke`
- RZ zigzag segment deposition + physical metric + escape: `RreaSegmentDepositionRZ.H`; pin: analytic case suite + the production continuity gate
- CIC thinning + global survivor/candidate transaction plan (each support gives up the same fraction of its removable capacity): `RreaCicConservingResample.H` + `RreaCouplingPopulation.cpp`, with the GPU support records `PopulationRecordBuilder` (`RreaPopulationRecord.H`); pin: `rrea_cic_resample_smoke` + `rrea_population_gpu_smoke` + one-/four-rank continuity fixture; capacity/floor in the reduced CSV
- Population action selector + adaptive split (each split moves exactly half the parent weight to a clone, so $E[w]$ and the deposited cell charges are preserved): `RreaAdaptiveResampleAction` / `RreaAdaptiveResampleSplitOutcome` (`RreaRng.H`), applied in `RreaCouplingPopulation.cpp`; pin: `rrea_rng_endpoint_smoke`
- RZ shell volume $V = \pi(r_{\mathrm{out}}^2 - r_{\mathrm{in}}^2)\Delta z$, both radii clamped to the axis: `rrea::rz_shell_volume` (`RreaConstants.H`), the one owner for the reduced observables, the event accumulator, the fluid face transport, the Maxwell metric and the video frame; pin: `rrea_event_accumulator_smoke` + `rrea_reduced_observables_smoke`
- Seed injection window/momentum: `RreaSeedSchedule.H`; pin: `rrea_seed_schedule_smoke`
- PARMA continuous cosmic-ray e^-/e^+ source (CSDA birth rate $Q_s(>E, z) = \phi_s(E, z)\, S(E)\, \chi(z)$, $q_s = -\mathrm{d}Q_s/\mathrm{d}E$, $\dot{N}_s = A \int Q_s(>E_{\min}, z)\, \mathrm{d}z$ per species with the ensemble drag S; PARMA at the run's own atmospheric depth; stationary bath $\phi_s/\beta c$ at $t = 0$; macros allocated $\propto \dot{N}_s$. Combined production writes the existing pulse (`write_seed_schedule`) first, then appends PARMA with disjoint macro indices. Pulse resolution uses `--macro-count`; background resolution uses the existing PARMA configuration independently. Extensions retain both initial components once, their weights and the original sampling interval, adding only future PARMA events. No old pulse-only checkpoint migration is provided. Continuous seeds do not establish multi-pulse recurrence; field evolution and feedback still determine it. Sato 2015/2016, non-commercial JAEA scope; fixed 2019-05-27 solar reference and the shared $e^{\pm}$ angular law are stated model limitations): `scripts/rrea_parma_source.py` (+ the thin Fortran evaluator `scripts/parma_electron_query.f90` against the vendored `rrea_parma_upstream/` model code and data; schedule `species_id` routes each event to its WarpX container in `InjectScheduledSeeds`); pin: `test_parma_source_physics.py` + `rrea_seed_schedule_smoke`
- Strict ceilings are memory backstops, not a physical process: a collective preflight in the coupling (in a unit-weight C&D run an abort is itself a result, electron count above five times the 276 reference)
- Plane-flux observable: `RreaPlaneFlux.H`; pin: transport-path smoke + report-side pins

- Event-source shell-volume normalization + off-box deposit routing (per-thread scratch FABs folded into the fields once per step by `FoldScratchIntoFields`, then beyond-ghost deposits routed to their exact owning cells in one collective flush): `RreaEventAccumulator.H` + `native_amrex_rrea/src/RreaEventAccumulator.cpp`; pin: `rrea_event_accumulator_smoke`
- Video-frame binning (the five display arrays resampled onto the fixed $n_r \times n_z$ grid on the same shell volume, then float32/shuffle/deflate): `RreaCouplingVideoFrame.cpp` + `RreaVideoFrameCodec.H`; pin: `scripts/test_driver_restart_safety.py` (the v4 stream contract)

Observables and analysis (Python unless noted)

- Unique energetic-electron first passage and weighted spatial tally (electrons): `RreaElectronCreation.H` (C++); pin: `rrea_electron_creation_smoke`, shared-history tests in `rrea_kinetic_engine_smoke`; annular-volume normalization (m^{-3}): `plot_rrea_electron_creation.py`, pinned by `test_delivery_physics.py`.
- Reduced field $E/N = |E|/(\mathrm{Td}\,\chi N_0)$: `reduced_field_td` (`RreaConstants.H`), used by the closure evaluation and the reduced observables; pin: `rrea_reduced_observables_smoke`
- Field-based reduced observables (E/N percentiles, field fractions, conduction-current extents, one region loop): `RreaReducedObservables.H` (C++); pin: `rrea_reduced_observables_smoke` vs analytic fields
- Fluid current moment and its z profile ($\sum J_z\,\mathrm{d}V$ of the step-averaged fluid face current Maxwell receives, so fluid plus net-chord equals `maxwell_jz_moment_A_m`): `MaxwellJzMomentProfile` (`RreaMaxwellTMRZ.H`); pin: `rrea_maxwell_tm_rz_smoke` (uniform-current moment)
- Species moments (M(t), dM/dz, energy census): coupling species sweeps (WarpX-bound); pin: `rrea_segment_deposition_smoke` pins deposited current moment $= qw\Delta r/\mathrm{d}t$ against the net chord
- Retarded Hertzian-dipole terms (D&C eq. 12 is the companion $\mathbf{B}$) + charge moment: `dipole_field_terms` / `cumulative_charge_moment`; pin: EFCM-null + LIP-step pins
- Finite-source line dipole: `finite_source_fields`; pin: runtime 1% conservation gate
- On-axis retarded surface kernel: `on_axis_longitudinal_fields` (`compute_rrea_em_radio.py`); pin: symmetry, causality and kernel pins in `test_delivery_physics.py`
- Off-axis Love-equivalence field: `off_axis_fields`; pin: interior static point charge vs exact Coulomb, plus symmetry/causality/convergence pins
- Shared retarded interpolation and sum: `retarded_surface_sum` (`rrea_retarded_sum.py`), $c_0 f(t - R/c) + c_1 \dot{f}(t - R/c)$; pin: analytic Coulomb/dipole and causal-front checks in `test_radio_physics.py`, and axis checks in `test_delivery_physics.py`

- Instrument responses (RC high-pass single owner, digitizer low-pass, boxcar single owner, Butterworth bands, spectra): `compute_rrea_radio_waveform.py` + multiband units; pin: DC-gain, Parseval and causal-Butterworth pins
- Resolved radio band (Yee phase error across the grid, azimuth quadrature): `resolved_max_hz` (`compute_rrea_em_radio.py`), caps every radio label; pin: exact Yee dispersion pin in `test_radio_physics.py`
- EFCM plate + LIP field mill: `efcm_plate_field` / `lip_vertical_field`; pin: exact-nadir-null + step-survival pins
- Optical yields and C-state branching: `fy337_ph_per_mev`, `photon_probability_337`, `quench_factor_777` in `compute_rrea_optical_emissions.py`; pin: absolute AIRFLY yield, branching and density limits in `test_optical_physics.py`. `native_pair_source_series` (pairs/s, m) is pinned by source-moment recovery and pair-count conservation under smoothing.
- Heterogeneous source-yield bounds and sampled swarm envelope: `source_yield_bounds`, `swarm_emission_envelope` (photons, ph/s); pin: positive spatial mixtures, nonmonotone rates, afterglow and missing-data checks in `test_optical_physics.py`
- Offline molecular rates: `phelps_collisions`, `excitation_rates` in `generate_rrea_fluid_closure_bundle.py` (eV, m^2, m^3/s); pin: source thresholds/units, elastic remainder, native process rate/active domain and tail integral in `test_optical_physics.py`.
- Cloud transport coupling: `rrea_cloud_scattering.py` (`scattered_flux`, config builder); pin: committed physical self-test + absolute-clock pin
- C&D avalanche integral $N_\lambda = \int \frac{E - E_{th}\rho}{7300\,\mathrm{kV}}\,dz$: `estimate_integrated_avalanche_lengths`; pin: closed-form slab pin in `test_cd_science.py`
- Signed plane multiplier + cohort deviations (both thresholds): report/analysis physics functions; pin: pinned in `test_cd_science.py`
- Ambient-field audit + Td identity: `audit_ambient_field` (`rrea_profiled_atmosphere.py`); pin: production-margin + identity pins
- Band series (population, not flux): `extract_rrea_band_series.py`; pin: analytic annular-volume and selected-band checks in `test_delivery_physics.py`

16 Glossary

Avalanche length λ Distance over which the runaway population multiplies by e; benchmarked against the Coleman–Dwyer formula under both threshold conventions (Section 12).

Relativistic feedback Dwyer's mechanism: backward-propagating bremsstrahlung photons and pair-produced positrons from one avalanche seed new avalanches at the source region; above a threshold field the discharge becomes self-sustaining. Emerges naturally here from tracked photon/positron transport.

Macroparticle / weight One computational particle standing for w real particles; all its interactions are sampled once and applied at weight w.

MultiFab AMReX's distributed mesh field (one physical quantity on all boxes of all ranks).

PIC Particle-in-cell: particles for the kinetic species, a mesh for fields, coupled by deposit/gather.

Optical depth $\int n\,\sigma\,\mathrm{d}\ell$ along a path; the expected number of interactions. Sampling "the next event at optical depth $-\ln u$" is the standard MC transport trick used per channel.

Demotion Converting a below-cutoff charged secondary from a would-be particle into fluid density sources (charge- and energy-conserving).

Density scaling Every table rate is rescaled at run time by the local ρ/ρ_0 (mean free paths $\propto \rho_0/\rho$, drags $\propto \rho/\rho_0$), from either a constant ratio or an altitude profile. The table reference, the RREA threshold reference and the Loschmidt reference for mobilities and E/N are explicitly distinguished; use the normalization for the quantity concerned (Sections 7.1 and 12).

Maxwell TM solve The supported finite-speed RZ field update for (E_r, E_z, B_θ) with kinetic and fluid face currents (Section 8.3).

Continuity/Gauss Material continuity is a fail-closed outer-step gate; the Gauss defect is a monitor. Neither applies a correction.

Schema-6 The versioned interaction-table generation shipped here: Geant4-anchored and semantic-validation checked.

Fail-closed Invalid inputs or failed invariants stop the operation; documented numerical limiters and recoverable append tails have explicit handling.

Ceiling A hard cap on macroparticle count per species. Under the production driver's `adaptive_resample_v1` it is a fail-closed backstop above the control band; under `strict_abort_v1` reaching it is a collective abort. Never a silent drop (§13).

Capture One production run of `scripts/run_rrea_profiled_video_capture.py` and the directory it writes (reduced CSV, frame stream, checkpoints and side-car series), which every offline pipeline in §10 reads.